MODULE : 5

3/10/19

FINITE IMPULSE RESPONSE FILTER

Finite impulse response filter:

→ A FIR filter has finite number of samples in its impulse response.

→ The output of FIR filter depends on present and past input values.

→ FIR filters can be used in data transmission, speech processing, correlation processing etc.

Characteristics of FIR filter:

→ The impulse response of FIR filter is of finite length.

→ The non-recursive FIR filter are always stable

→ The undesirable phase distortion resulted due to non-linear characteristics of the frequency response can be removed by FIR filter.

→ Quantization noise is also eliminated by FIR filter.

Advantages of FIR filter :-

→ FIR filters are stable as it is not recursive.

→ FIR filters can be realised in both recursive and non-recursive structures.

→ FIR filter can have exact linear phase

→ FIR filters can be efficiently realised in hardware

→ Low sensitivity to quantisation noise.

* **Disadvantage :**

→ Practical realisation is complex.

→ The realisation of FIR - filter which has long duration impulse response requires larger amount of processing.

→ Compare to IIR filter, appreciably higher order filter is required to achieve specific magnitude response, which requires more filter coefficients.

The various structures for the realisation of FIR filters are :

i) direct form realisation
ii) frequency sampling structure
iii) Linear phase structure
iv) Lattice structure.

The major factors that influence the choice of specific realization are

→ i) Computational complexity.

↳ it refers to number of arithmetic operations required to compute an output of the system.

ii) Memory requirement :

It refers to number of memory locations required to store system parameters and intermediate computed values.

iii) Finite word length effect :

It refers to the quantisation effect that are inherent in any digital implementation of the system.

→ Difference between FIR and IIR filters:

FIR filters	IIR filters
→ These are non-recursive	→ These are recursive
→ These filters produce zeros.	→ These filters produce poles.
→ In signal processing, FIR filter is a filter whose impulse response is of finite duration, because it settles to zero in finite time.	→ This is in contrast to IIR, it may have internal feedback and may continue to respond indefinitely.
→ Filters combining both past inputs and past outputs can produce both poles and zeros.	→ IIR filters are definitely difficult to control and have no particular phase.
→ FIR filters can be discreet-time or continuous-time and digital or analog.	→ IIR is derived from analog.
→ FIR filters are dependent upon linear phase characteristics.	→ IIR filters are used for applications that are not linear.
→ FIR is always stable.	→ IIR can be unstable.
→ FIR is dependent on input only.	→ IIR are dependent on both input and output.
→ FIR's require more memory.	→ IIR requires less memory.

The filter's response settles to zero in finite time.
FIR filters contain as many poles as they have zeros, but all of the poles are located at the origin, z=0 because all of the poles are located inside the unit circle, the FIR filter is intrinsically stable.

Scanned with CamScanner

→ Structure for the realisation of discreet time system. 4/10/19

Let us consider LTI system, characterized by the general linear constant, coefficient difference equation given by,

$$y(n) = -\sum_{k=1}^{N} a_k\, y(n-k) + \sum_{k=0}^{N-1} b_k\, x(n-k) \quad -①$$

then the transfer function is given by,

$$H(z) = \frac{Y(z)}{X(z)}$$

$$Y(n) + \sum_{k=1}^{N} a_k\, y(n-k) = \sum_{k=0}^{N-1} b_k\, x(n-k)$$

$$Y(z) + \sum_{k=1}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{N-1} b_k z^{-k} X(z).$$

$$Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = \sum_{k=0}^{N-1} b_k z^{-k} \cdot X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{N-1} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} \quad -②.$$

The focus is to realize eqn ① and ② using different structures.

STRUCTURES FOR FIR FILTER :

The FIR filter is described by the difference eqn,

$$y(n) = \sum_{k=0}^{M-1} b_k\, x(n-k) \quad , \text{ and}$$

$$H(z) = \sum_{k=0}^{M-1} b_k z^{-k}$$

The unit sample response of the FIR filter is identical to the coefficient, that is,

$$h(n) = \begin{cases} b_n & 0 \le n \le M-1 \\ 0 & \text{otherwise.} \end{cases}$$

The direct form realisation is also called transversal, tapped delay line filter.

→ **Direct form Realisation of FIR filter:**

(Transverse / Tapped delay line)

The difference equation of FIR filter is given by,

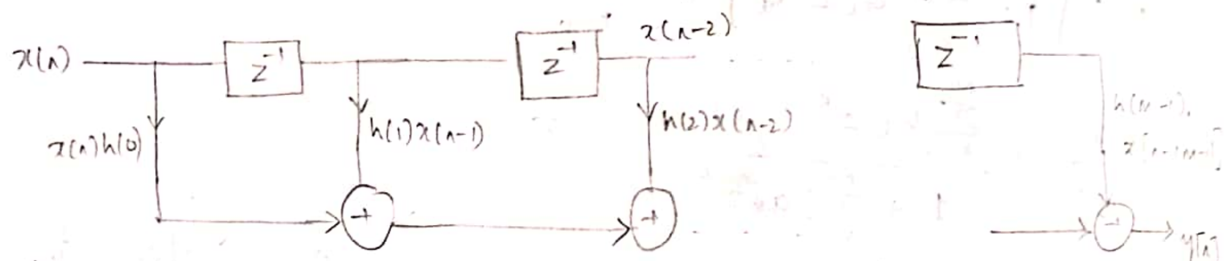
$$y(n) = \sum_{k=0}^{M-1} b_k x(n-1).$$

w.k.t, $h(n) = \begin{cases} b_n & , 0 < n \le M-1 \\ 0 & \text{otherwise} \end{cases}$

substituting for b_k,

$$y[n] = \sum_{k=0}^{M-1} h(k)\, x(n-k).$$

$$y[n] = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots + h(M-1)x[n-(M-1)]$$



This structure requires $(M-1)$ memory location for storing $(M-1)$ previous i/ps and no. of opns performed are M → Multpcn, $(M-1)$ → addtns per o/p.

Problem:

1. Given $H(z) = 1 + 2z^{-1} + 3z^{-2} - 4z^{-3} + 5z^{-4}$, draw direct form realization.

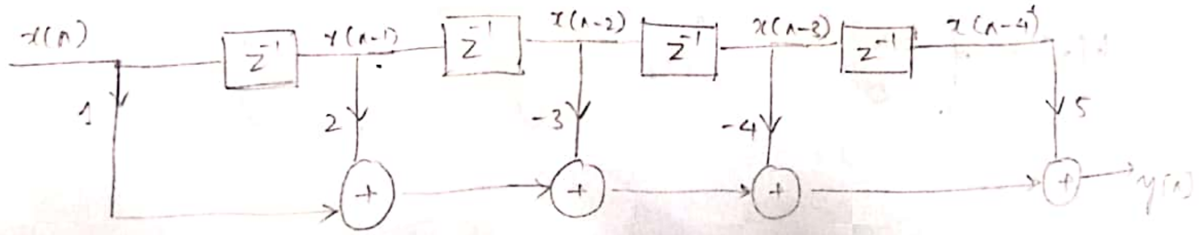
Soln:

$$H(z) = \frac{Y(z)}{X(z)} = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3} + 5z^{-4}$$

$$Y(z) = X(z) + 2z^{-1}X(z) - 3z^{-2}X(z) - 4z^{-3}X(z) + 5z^{-4}X(z)$$

↓ IZT

$$y(n) = x(n) + 2x(n-1) - 3x(n-2) - 4x(n-3) + 5x(n-4).$$

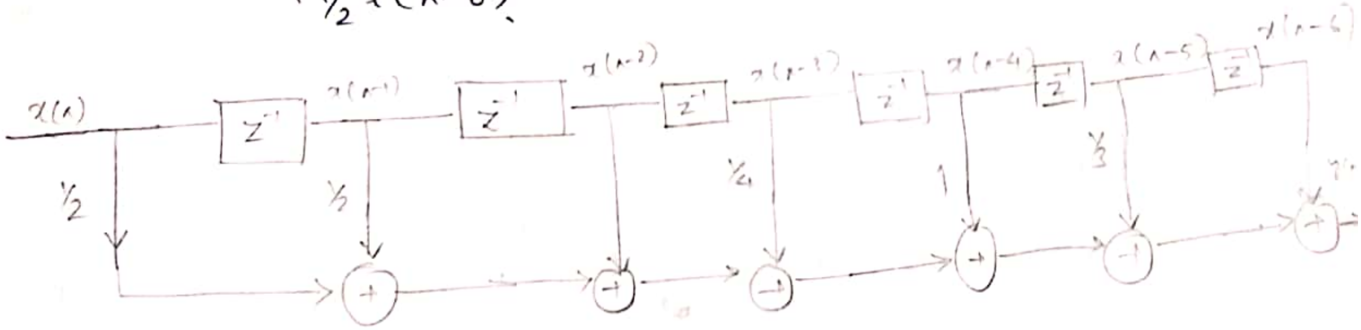


2) $H(z) = \frac{1}{2} + \frac{1}{2}z^{-1} + z^{-2} + \frac{1}{4}z^{-3} + z^{-4} + \frac{1}{3}z^{-5} + \frac{1}{2}z^{-6}$.

$$\frac{Y(z)}{X(z)} = \frac{1}{2} + \frac{1}{2}z^{-1} + z^{-2} + \frac{1}{4}z^{-3} + z^{-4} + \frac{1}{3}z^{-5} + \frac{1}{2}z^{-6}.$$

$$Y(z) = \frac{1}{2}X(z) + \frac{1}{2}z^{-1}X(z) + z^{-2}X(z) + \frac{1}{4}z^{-3}X(z) + z^{-4}X(z) + \frac{1}{3}z^{-5}X(z) + \frac{1}{2}z^{-6}X(z)$$

$$y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1) + x(n-2) + \frac{1}{4}x(n-3) + x(n-4) + \frac{1}{3}x(n-5) + \frac{1}{2}x(n-6).$$



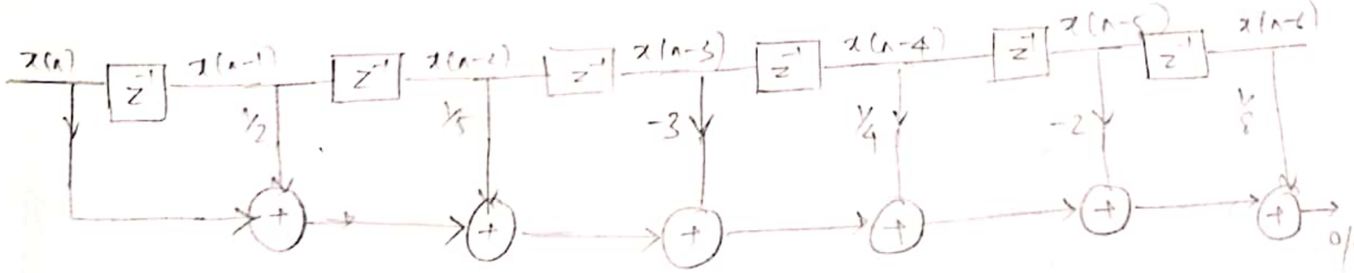
3). $H(z) = 1 + \frac{1}{2}z^{-1} + \frac{1}{5}z^{-2} - 3z^{-3} + \frac{1}{4}z^{-4} - 2z^{-5} + \frac{1}{8}z^{-6}$.

$$\frac{Y(z)}{X(z)} = 1 + \frac{1}{2}z^{-1} + \frac{1}{5}z^{-2} - 3z^{-3} + \frac{1}{4}z^{-4} - 2z^{-5} + \frac{1}{8}z^{-6}.$$

$$Y(z) = X(z) + \frac{1}{2}z^{-1}X(z) + \frac{1}{5}z^{-2}X(z) - 3z^{-3}X(z) + \frac{1}{4}z^{-4}X(z) - 2z^{-5}X(z) + \frac{1}{8}X(z)z^{-6}.$$

$\Downarrow$ I.Z.T

$$y[n] = x(n) + \frac{1}{2}x(n-1) + \frac{1}{5}x(n-2) - 3x(n-3) + \frac{1}{4}x(n-4) - 2x(n-5) + \frac{1}{8}x(n-6)$$



Scanned with CamScanner

* Linear phase FIR filter Structure :

If the unit sample response of a FIR filter satisfies $h(n) = h(N-1-n)$, then the filter is said to have linear phase.

i.e., It has symmetric response about its origin.

For FIR filter,

$$H(z) = \sum_{n=0}^{N-1} h(n).z^{-n}.$$

<u>Case i)</u> for <u>N = even</u>,

$$H(z) = \sum_{n=0}^{N/2-1} h(n).z^{-n} + \sum_{n=N/2}^{N-1} h(n) z^{-n}.$$

if $n_{new} = N-1-n$.

If $n = N/2$, $\quad n_{new} = N/2 - 1$.

If $n = N-1$, $\quad n_{new} = 0$.

$$H(z) = \sum_{n=0}^{N/2-1} h(n) z^{-n} + \sum_{n=N/2-1}^{0} h(N-1-n) z^{-(N-1-n)}$$

w.k.t, $h(N-1-n) = h(n)$.

$$H(z) = \sum_{n=0}^{N/2-1} h(n).z^{-n} + \sum_{n=0}^{N/2-1} h(n).z^{-(N-1-n)}$$

$$\frac{Y(z)}{X(z)} = \sum_{n=0}^{N/2-1} h(n) \left[z^{-n} + z^{-(N-1-n)}\right]$$

$$Y(z) = \sum_{n=0}^{N/2-1} h(n) \left[z^{-n} + z^{-(N-1-n)}\right] X(z)$$

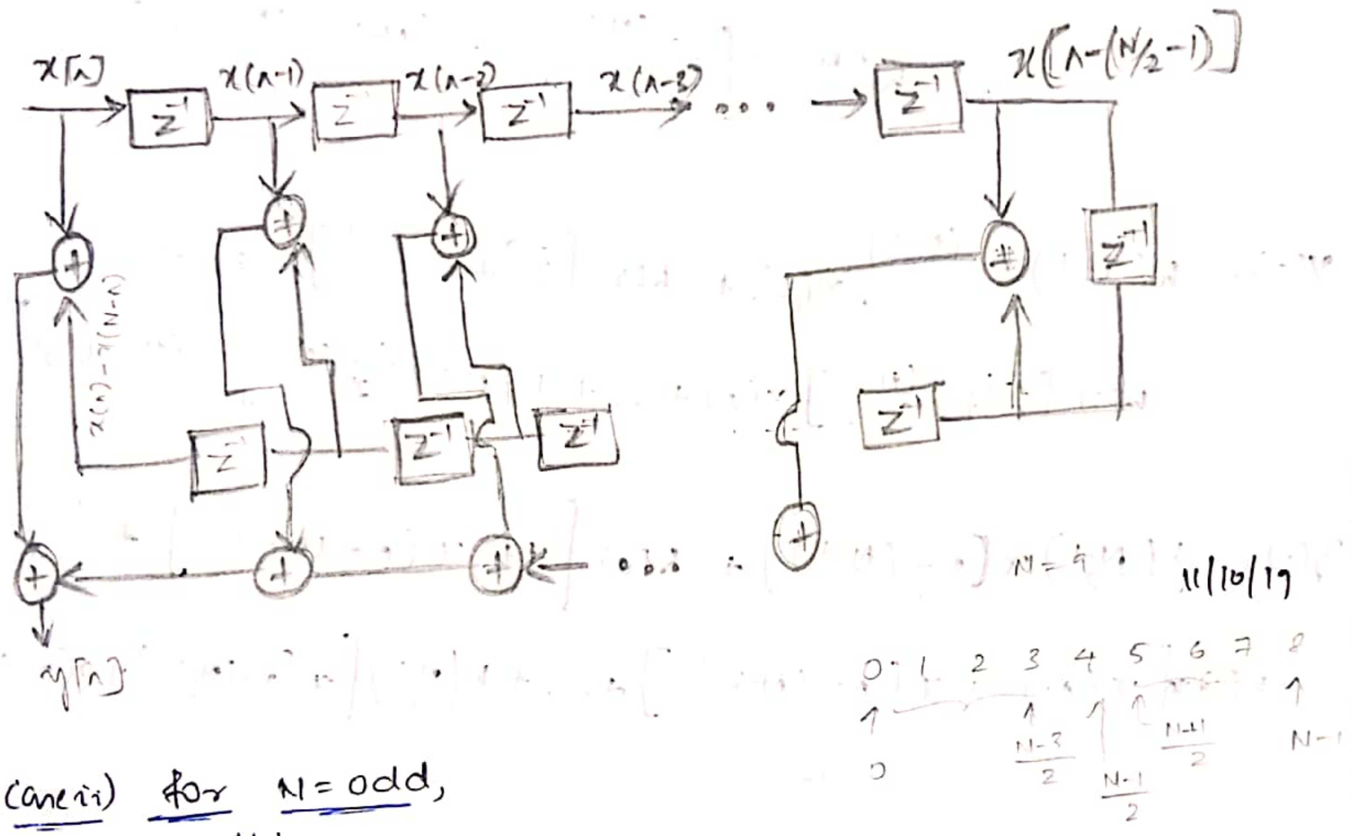
$$Y(z) = h(0)\left[z^{-0} + z^{-(N-1-0)}\right]X(z) + h(1)\left[z^{-1} + z^{-(N-1-1)}\right]X(z) + \dots$$
$$h\left(\tfrac{N}{2}-1\right)\left[z^{-(N/2-1)} + z^{-(N/2-1-N/2+1)}\right]$$

$$Y(z) = h(0)\left[1 + z^{-(N-1)}\right]X(z) + h(1)\left[z^{-1} + z^{-(N-2)}\right] + \dots +$$
$$h\left(\tfrac{N}{2}-1\right)\left[z^{-(N/2-1)} + z^{-(N/2)}\right] X(z)$$

Inverse z-transform,

$$y[n] = h(0)x[n] + h(0)x[n-(N-1)] + h(1)x(n-1) + h(1)x[n-(N-2)] + \dots$$
$$\dots + h(N/2-1)\,x[N/2-1] + h(N/2-1)\,x[n-\{N/2\}]$$

$$y[n] = h(0)\left[x[n] + x[n-(N-1)]\right] + h(1)\left[x(n-1) + x[n-(N-2)]\right] + \dots$$
$$+ h(N/2-1)\left[x[n-N/2-1] + x[n-N/2]\right].$$



11/10/19

case ii) for N = odd,

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n}$$

let

$$H(z) = \sum_{n=0}^{N-3} h(n) z^{-n} + \underline{h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)}} + \sum_{n=\frac{N-1}{2}}^{N-1} h(n) z^{-n}.$$

let $n_{new} = N-1-n$

If $n = \frac{N+1}{2}$, $n_{new} = N-1-\frac{N+1}{2} = \frac{N-3}{2}$

If $n = N-1$, $n_{new} = 0$

$$H(z) = \sum_{n=0}^{N-3/2} h(n) z^{-n} + h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N-3}{2}}^{0} h(N-1-n)\, z^{-(N-1-n)}$$

w.k.t, $h(N-1-n) = h(n)$

$$H(z) = h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n).z^{-n} + \sum_{n=0}^{\frac{N-3}{2}} h(n).z^{-(N-1-n)}$$

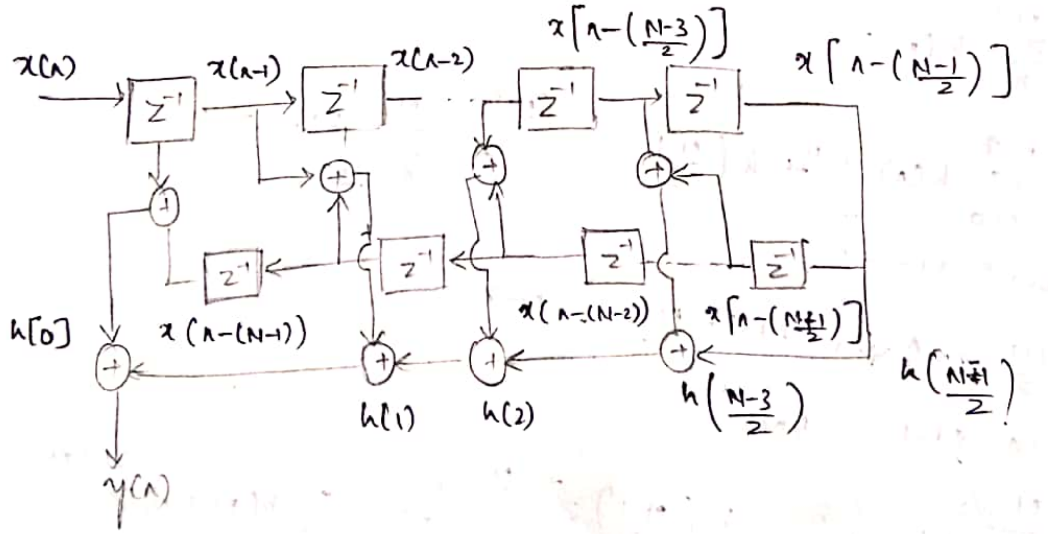
$$H(z) = h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n)\left[z^{-n} + z^{-(N-1-n)}\right]$$

$$\frac{Y(z)}{X(z)} = H(z)$$

$$= h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + h(0)\left[z^{-0} + z^{-(N-1)}\right] + h(1)\left[z^{-1} + z^{-(N-2)}\right] + \dots h\left(\frac{N-3}{2}\right)\left[z^{-\left(\frac{N-3}{2}\right)} + z^{-\left(N-1-\left(\frac{N-3}{2}\right)\right)}\right]$$

$$Y(z) = h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)}.X(z) + h(0)\left[z^{-0} + z^{-(N-1)}\right]X(z) + h(1)\left[z^{-1} + z^{-(N-2)}\right]X(z) + \dots + h\left(\frac{N-3}{2}\right)\left[z^{-\left(\frac{N-3}{2}\right)} + z^{-\left(N-1-\left(\frac{N-3}{2}\right)\right)}\right]$$

$$y(n) = h\left(\frac{N-1}{2}\right) x\left[n - \left(\frac{N-1}{2}\right)\right] + h(0)\left[1 + x(n-(N-1))\right] + h(1)\left[x(n-1) + x(n-(N-2))\right] + \dots + h\left(\frac{N-3}{2}\right)\left[x\left(n - \left(\frac{N-3}{2}\right)\right) + x(n-\dots\right]$$



Problems :)

1. Determine $H(z)$ of an FIR filter to implement, $h(n) = \delta(n) + 2\delta(n-1) + \delta(n-2)$. Draw the linear phase FIR filter for $N = 3$.

Soln: $h(n) = \delta(n) + 2\delta(n-1) + \delta(n-2)$

$\downarrow$

$$H(z) = 1 + 2z^{-1} + z^{-2}$$

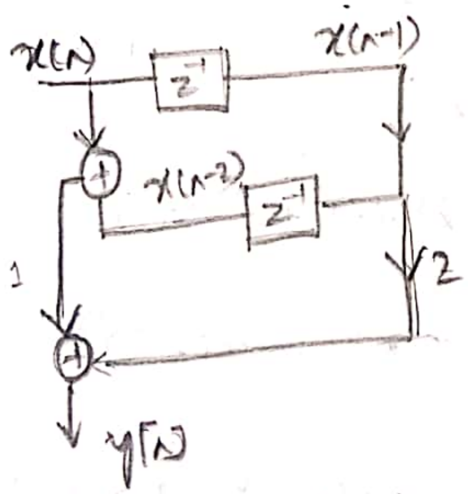
$$\frac{Y(z)}{X(z)} = 1 + 2z^{-1} + z^{-2}$$

$$Y(z) = X(z) + 2z^{-1}X(z) + z^{-2}X(z)$$

$\downarrow$ IZT

$$y(n) = x(n) + 2x(n-1) + x(n-2)$$

$$y(n) = 1[x(n) + x(n-2)] + 2x(n-1)$$



2. $h(n) = \delta(n) + \frac{1}{4}\delta(n-1) + \frac{1}{8}\delta(n-2) + \delta(n-4) - \frac{1}{4}\delta(n-3)$

Soln: $H(z) = 1 + \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2} + z^{-4} - \frac{1}{4}z^{-3}$

$\downarrow$

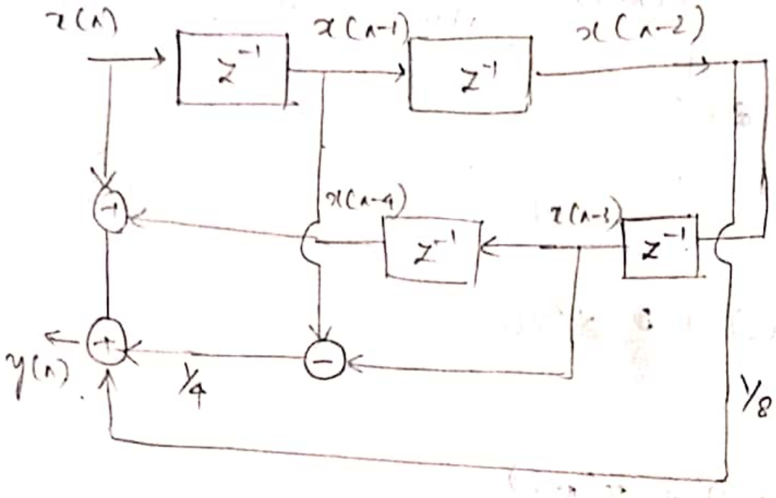
$$Y(z) = X(z) + \tfrac{1}{4}z^{-1}X(z) + \tfrac{1}{8}z^{-2}X(z) + z^{-4}X(z) - \tfrac{1}{4}z^{-3}X(z)$$

$\downarrow$ IZT

$$y(n) = x(n) + \tfrac{1}{4}x(n-1) + \tfrac{1}{8}x(n-2) + x(n-4) - \tfrac{1}{4}x(n-3)$$

$$y(n) = x(n) + \frac{1}{4}x(n-1) + \frac{1}{8}x(n-2) - \frac{1}{4}x(n-3) + x(n-4)$$

$$y(n) = \{x(n) + x(n-4)\} + \frac{1}{4}\{x(n-1) - x(n-3)\} + \frac{1}{8}x(n-2)$$



3. $h(n) = \delta(n) + \frac{1}{2}\delta(n-1) - \frac{1}{4}\delta(n-2) + \frac{1}{2}\delta(n-3) + \delta(n-4)$

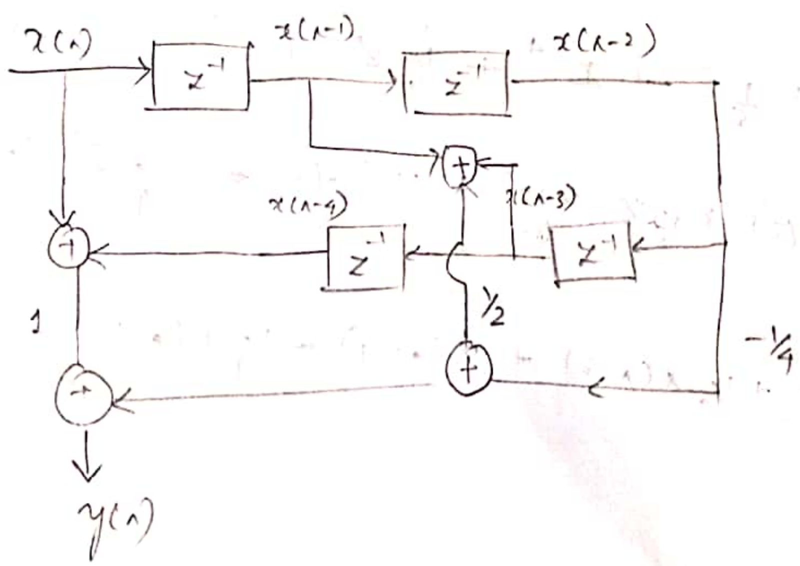
$$H(z) = 1 + \frac{1}{2}z^{-1} - \frac{1}{4}z^{-2} + \frac{1}{2}z^{-3} + z^{-4}$$

$$Y(z) = X(z) + \frac{1}{2}z^{-1}X(z) - \frac{1}{4}z^{-2}X(z) + \frac{1}{2}z^{-3}X(z) + z^{-4}X(z)$$

$\downarrow$ IZT

$$y(n) = x(n) + \frac{1}{2}x(n-1) - \frac{1}{4}x(n-2) + \frac{1}{2}x(n-3) + x(n-4)$$

$$y(n) = \{x(n) + x(n-4)\} + \frac{1}{2}\{x(n-1) + x(n-3)\} - \frac{1}{4}x(n-2)$$



4) $H(z) = \frac{2}{3} z^{1} + 1 + \frac{2}{3} z^{-1}$

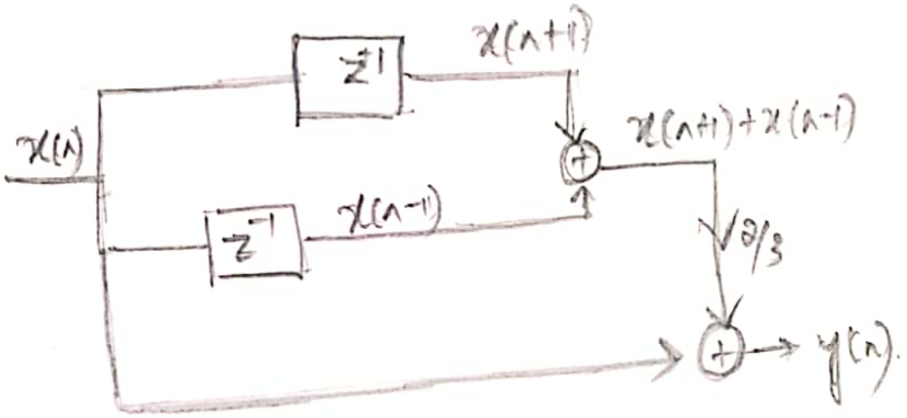
$$\frac{Y(z)}{X(z)} = \frac{2}{3} z^{1} + 1 + \frac{2}{3} z^{-1}$$

$$Y(z) = \frac{2}{3} z^{1} X(z) + X(z) + \frac{2}{3} z^{-1} X(z)$$

$$\downarrow$$

$$y(n) = \frac{2}{3} x(n+1) + x(n) + \frac{2}{3} x(n-1)$$

$$y(n) = x(n) + \frac{2}{3} [x(n+1) + x(n-1)]$$

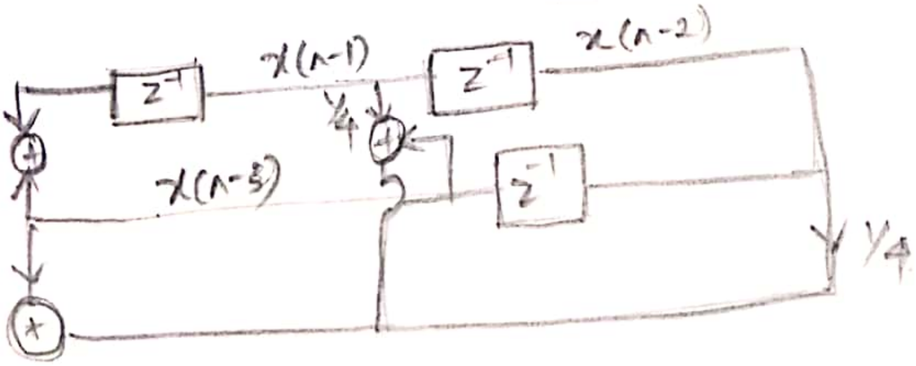


5) $H(z) = 1 + \frac{z^{-1}}{4} + \frac{z^{-2}}{4} + z^{-3}$

$$Y(z) = X(z) + \frac{z^{-1}}{4} X(z) + \frac{z^{-2}}{4} X(z) + z^{-3} X(z)$$

$$Y(z) = x(n) + \frac{1}{4} x(n-1) + \frac{1}{4} x(\overset{n-2}{n}) + \cancel{\frac{1}{4}} x(n-3)$$

$$Y(n) = \{x(n) + x(n-3)\} + \frac{1}{4}\{x(n-1) + x(n-2)\}$$



11/10/19

→ Frequency Sampling Structure:

- The frequency sampling realisation characterises the filter parameter based on the values of desired frequency response instead of impulse response $h(n)$.

- To derive frequency sampling structure, desired frequencies are sampled uniformly. i.e,

$$\omega_k = \frac{2\pi}{M}(n+\alpha) \quad ; \alpha = 0 \text{ or } 1/2 .$$

The frequency response is given by,

$$H(\omega) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n}$$

Sampling at $\frac{2\pi n(k+\alpha)}{M}$, we get.

$$H(k+\alpha) = \sum_{n=0}^{M-1} h(n) e^{\frac{-j2\pi(k+\alpha)n}{M}}$$

The set of value $H(k+\alpha)$ are the samples of $H(\omega)$ if $\alpha = 0$ in above equation, then, $H(k)$ corresponds to M point DFT of $h(n)$

∴ IDFT is,

$$h(n) = \frac{1}{M} \sum_{n=0}^{M-1} H(k+\alpha) e^{\frac{j2\pi(k+\alpha)n}{M}} \quad — ①$$

$$H(z) = \frac{1}{M} \sum_{k=0}^{M-1} H(k+\alpha) e^{\frac{j2\pi(k+\alpha)n}{M}}$$

$$\cancel{\frac{1}{M} \sum_{k=0}^{M-1} H(k+\alpha)}$$

$$\cancel{\forall k, e^{j2\pi k}} = 1$$

$$H(z) = \sum_{n=0}^{M-1} h(n) z^{-n} \quad — ②$$

substitute ① in ②

$$H(z) = \sum_{n=0}^{M-1} \left[\frac{1}{M} \sum_{k=0}^{M-1} H(k+\alpha) e^{j\frac{2\pi(k+\alpha)n}{M}} \right] z^{-n}.$$

$$\frac{1}{M}\sum_{n=0}^{M-1} H(k+\alpha) \sum_{n=0}^{M-1} \left[e^{\frac{j2\pi(k+\alpha)}{M}} z^{-1}\right]^n$$

w.k.t

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$$

$$H(z) = \frac{1}{M}\sum_{k=0}^{M-1} H(k+\alpha)\left[\frac{1-e^{\frac{j2\pi(k+\alpha)M}{M}} z^{-M}}{1-e^{\frac{j2\pi(k+\alpha)}{M}} z^{-1}}\right]$$

$$\frac{1}{M}\sum_{k=0}^{M-1} H(k+\alpha)\left[\frac{1-e^{j2\pi k}\cdot e^{j2\pi\alpha} z^{-m}}{1-e^{\frac{j2\pi(k+\alpha)}{m}} z^{-1}}\right]$$

as k, $e^{j2\pi k} = 1$.

$$H(z) = \frac{1}{M}\sum_{k=0}^{M-1} H(k+\alpha)\left[\frac{1-e^{j2\pi\alpha} \cdot z^{-M}}{1-e^{\frac{j2\pi(k+\alpha)}{M}} z^{-1}}\right]$$

$$= \frac{1-e^{j2\pi\alpha} \cdot z^{-m}}{M} \sum_{k=0}^{M-1} \frac{H(k+\alpha)}{1-e^{\frac{j2\pi(k+\alpha)}{M}} z^{-1}}$$

$$H(z) = H_1(z) \cdot H_2(z)$$

$$H_1(z) = \frac{1-e^{j2\pi z} \cdot z^{-m}}{M}.$$

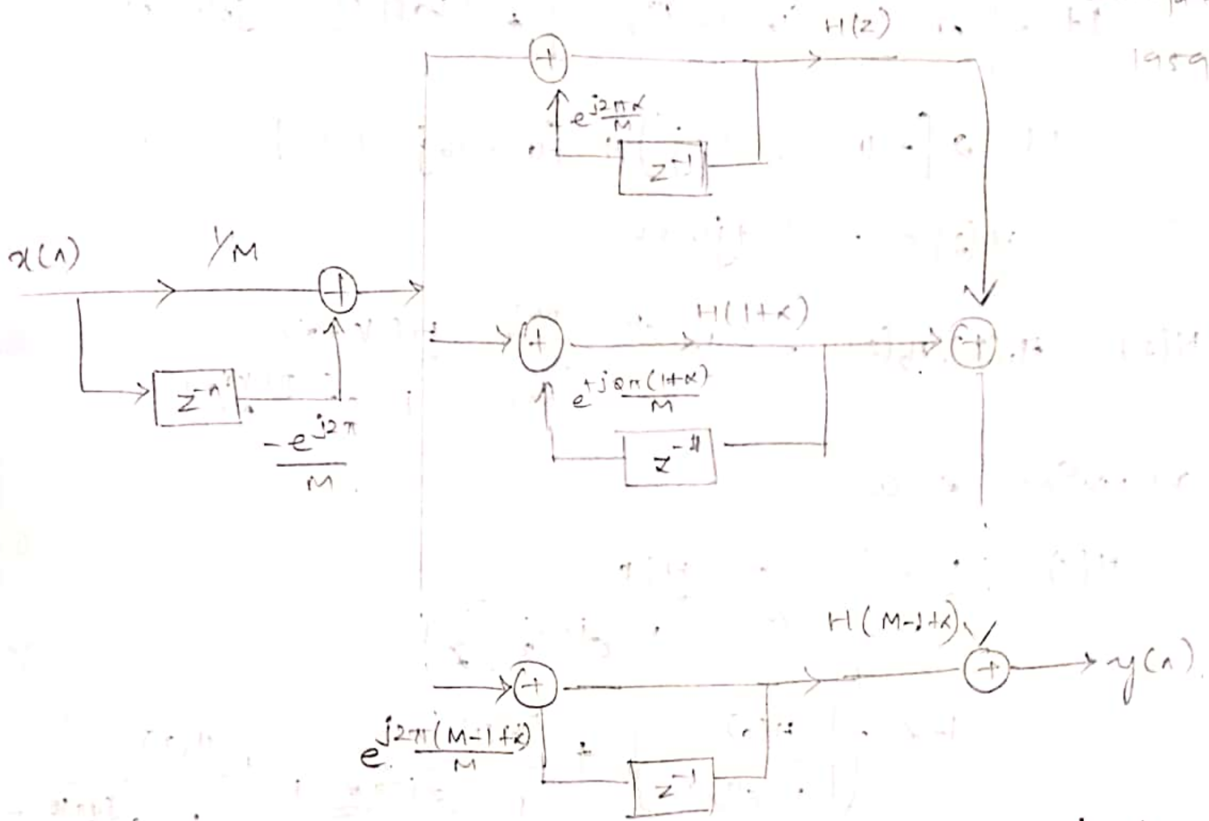
If $\alpha = 0$, $H_1(z) = \frac{1-z^{-M}}{M}$

If $\alpha = 1/2$, $H_1(z) = \frac{1+z^{-M}}{M}$

$$H_2(z) = \sum_{k=0}^{M-1} \frac{H(k+\alpha)}{1-e^{\frac{j2\pi(k-\alpha)}{M}} \cdot z^{-1}}$$

$$H(z) = H_1(z) \cdot H_2(z)$$

$$\frac{1-z^{-m}}{M} \sum_{k=0}^{M-1} \frac{H(k+\alpha)}{1-e^{\frac{j2\pi(k-\alpha)}{M}} \cdot z^{-1}}.$$



15/10/19.

Determine transfer function of FIR filter to implement $h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$ using frequency sampling technique.

Soln: $h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$ If $m-1=2$, $M=3$.

$h(0)\delta[n] + h(1)\delta[n-1] + h(2).\delta[n-2]$

$h(0)=1,\quad h(1)=2,\quad h(2)=1$

$$H(k) = \sum_{n=0}^{M-1} h(n)\, e^{-j\frac{2\pi kn}{N}} \qquad k=0..M-1$$

$$\sum_{n=0}^{2} h(n)e^{-j\frac{2\pi kn}{3}} \qquad k=0..2$$

$$h(0)e^0 + h(1)\,e^{-j\frac{2\pi k}{3}} + h(2)e^{-j\frac{4\pi k}{3}}$$

$$H(k) = 1 + 2e^{-j\frac{2\pi k}{3}} + e^{-j\frac{4\pi k}{3}}$$

If $k=0$, $H(0) = 1+2e^0+e^0 = 1+2+1 = 4.$

If $k=1$, $H(1) = 1+2e^{-j\frac{2\pi}{3}} + e^{-j\frac{4\pi}{3}}$

$$H(1) = 1 + 2\left[\cos\frac{2\pi}{3} - j\sin\frac{2\pi}{3}\right] + \left[\cos\frac{4\pi}{3} - j\sin\frac{4\pi}{3}\right]$$

$$1 + 2\left[-\tfrac{1}{2} - j\sqrt{3}/2\right] + \left[-\tfrac{1}{2} + j\sqrt{3}/2\right]$$

$$H(1) = -0.5 - j\sqrt{3}/2$$

If $k=2$, $H(2) = 1+2e^{-j4\pi/3} + e^{+j8\pi/3}$

$$1+2\left[\cos\left(\frac{4\pi}{3}\right) - j\sin\left(\frac{4\pi}{3}\right)\right] + \left[\cos\left(\frac{8\pi}{3}\right) - j\sin\left(\frac{8\pi}{3}\right)\right]$$

$$1+2\left[-0.5-0.866j\right] + \left[0.866j - 0.5\right]$$

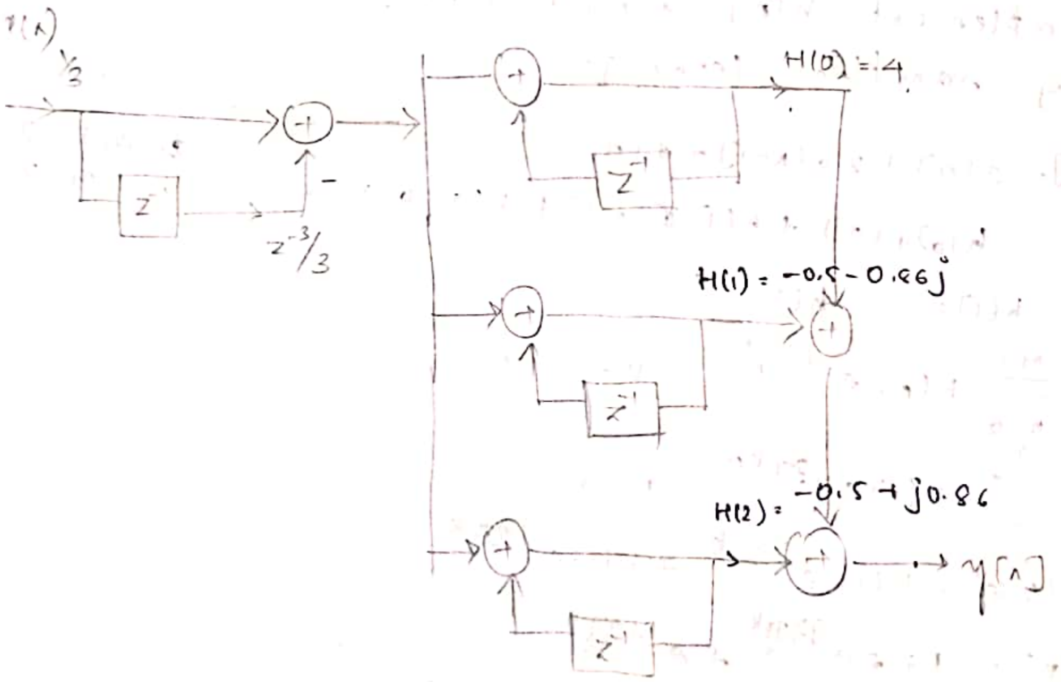
$$H(2) = -0.5 + j0.86$$

$$H(z) = H_1(z).H_2(z) = \frac{1-z^{-M}}{M}\sum_{k=0}^{M-1}\frac{H(k+\alpha)}{1-e^{j2\pi(k+\alpha)/M}.z^{-1}}$$

assuming $\alpha = 0$,

$$H(z) = \frac{1-z^{-3}}{3}\sum_{k=0}^{2}\frac{H(k)}{1-e^{j2\pi k/3}.z^{-1}}$$

$$\underbrace{\frac{1-z^{-3}}{3}}_{H_1(z)}\left\{\left[\frac{H(0)}{1-e^{0}z^{-1}}\right] + \underbrace{\frac{H(1)}{1-e^{+j2\pi/3}z^{-1}} + \frac{H(2)}{1-e^{j4\pi/3}.z^{-1}}}_{H_2(z)}\right\}$$



→ Lattice structure for FIR filter.

- Lattice structures are used for the different application in digital speech processing in the implementation of adaptive filter.
- Let the system function of m^{th} order FIR filter be represented by a polynomial $A_m(z)$

$$H_m(z) = A_m(z), \text{ where } A_m(z) = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k}$$

$$H_m(z) = A_m(z)$$

$$A_m(z) = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k}$$

$$\frac{Y(z)}{X(z)} = H_m(z) = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k}$$

$$Y(z) = X(z) + \sum_{k=1}^{m} \alpha_m(k) z^{-k} X(z)$$

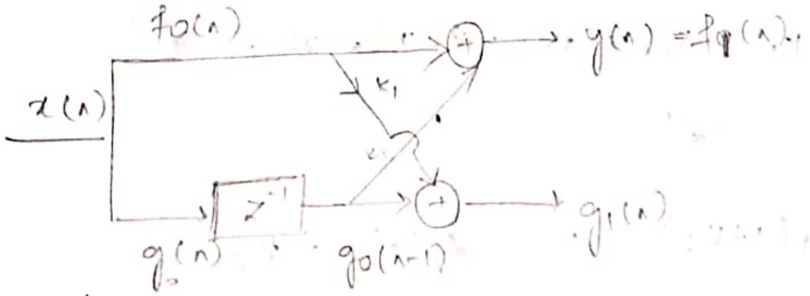
$\downarrow$ IZT

$$y(n) = x(n) + \sum_{k=1}^{m} \alpha_m(k)\, x(n-k)$$

case i) $m = 1$

$$y(n) = x(n) + \sum_{k=1}^{1} \alpha_1(k)\, x(n-k)$$

$$\left.\begin{aligned} &x(n) + \alpha_1(1)\, x(n-1) \\ &x(n) + K_1\, x(n-1) \end{aligned}\right\} \quad K_1 = \alpha_1(1)$$



$$f_0(n) = g_0(n) = x(n)$$

$$g_0(n-1) = x(n-1)$$

$$y(n) = f_1(n) = f_0(n) + K_1\, g_0(n-1)$$

$$g_1(n) = K_1 f_0(n) + g_0(n-1)$$

$$y(n) = f_0(n) = x(n) + K_1[x(n-1)]$$

$g_1[n] = k_1 x(n) + x(n-1)$.

Case (iii) $M = 2$

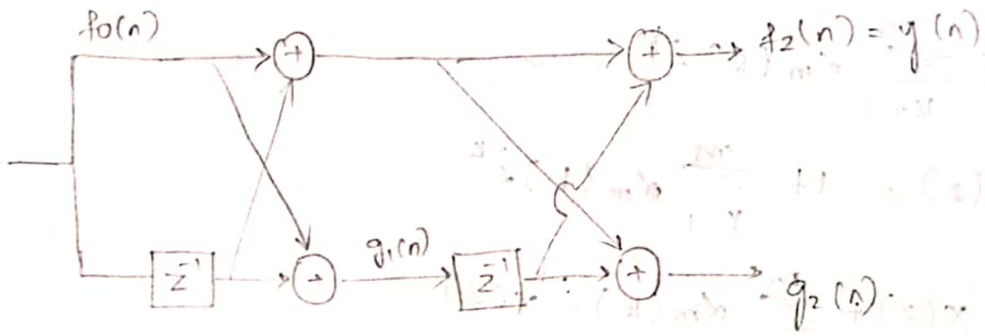
$$y(n) = x(n) + \sum_{k=1}^{2} \alpha_2(n)\, x(n-k)$$

$$y(n) = x(n) + \alpha_2(1) + x(n-1) + \alpha_2(2)\, x(n-2) \quad \text{—} (*)$$

$f_1(n) = x(n) + k_1 x(n-1)$ — (1)

$g_1(n) = k_1 x(n) + x(n-1)$ — (2)

$y(n) = x(n) + \alpha_2(1)\, x(n-1) + \alpha_2(2).\, x(n-2) \rightarrow (*)$



$y(n) = f_2(n) = f_1(n) + k_2\, g_1(n-1)$ — (3).

$y_2(n) = k_2 f_1(n) + g_1(n-1)$ — (4)

From (2),

$g_1(n-1) = k_1 x(n-1) + x(n-2) \rightarrow$ (5).

Subs, eqn (1) and (5) in (3),

$$y(n) = x(n) + k_1 x(n-1) + k_1 k_2 x(n-1) + k_2 x(n-2)$$

$$x(n) + k_1(1+k_2)\, x(n-1) + k_2\, x(n-2) \quad \text{—} (*)'$$

Comparing (*) and (*)'

$\alpha_2(1) = k_1(1+k_2)$; $\alpha_2(2) = k_2$

$$k_1 = \frac{\alpha_2(1)}{1+k_2}$$; k_1 and k_2 are

k_1 and k_2 are lattice coefficients.

→ Thus the reflection coefficient k_1 and k_2 of the lattice filter can be obtained ~~by~~ from by the coefficient $\alpha_m(k)$ or $a_m(k)$ of the direct form realisation

→ The lattice structure of m^{th} order can be obtained from the direct form coefficients by

$$K_m = a_m(m)$$

Conversion of direct form coefficient to lattice coefficient is done by using the expression

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m)\, a_m(m-i)}{1 - K_m^2}$$

Solving problem:

1. A FIR filter is given by
$y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$. Draw the lattice structure. {here M = 1 to M}.

Soln: $y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$

$\downarrow$ Z.T

$$Y(z) = X(z) + \frac{2}{3}z^{-1}X(z) + \frac{3}{4}z^{-2}X(z) + \frac{1}{3}z^{-3}X(z).$$

$$\frac{Y(z)}{X(z)} = 1 + \frac{2}{3}z^{-1} + \frac{3}{4}z^{-2} + \frac{1}{3}z^{-3}. \quad — ①$$

$$H_m(z) = A_m(z) = 1 + \sum_{k=1}^{m} a_m(k).z^{-k}$$

$$H_m(z) = 1 + \sum_{k=1}^{3} a_3(k).z^{-k}$$

$$1 + a_3(1).z^{-1} + a_3(2).z^{-2} + a_3(3)\,z^{-3} \quad — ②.$$

comp ① & ②,

$a_3(1) = 2/5$, $a_3(2) = 3/4$, $a_3(3) = 1/3$

$a_2(2) = k_2$

w.k.t $a_m(m) = k_m$

$a_1(1) = k_1$

$$a_3(3) = k_3 = 1/3$$

Case ii) $m=3$, $i=1$, $M-1 = 1, 2$

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m)\, a_m(m-i)}{1-K_m^2}$$

$M=3$, $i=1$

$$a_2(1) = \frac{a_3(1) - a_3(3).\, a_3(2)}{1-K_3^2}$$

$$a_2(1) = \frac{(2/5) - (1/3 * 3/4)}{1-(1/3)^2} = 0.168$$

$m=3$, $i=2$,

$$a_2(2) = \frac{a_3(2) - a_3(3).\, a_3(1)}{1-K_3^2}$$

$$\frac{(3/4) - (1/3) * (2/5)}{1-(1/3)^2} = 0.693$$

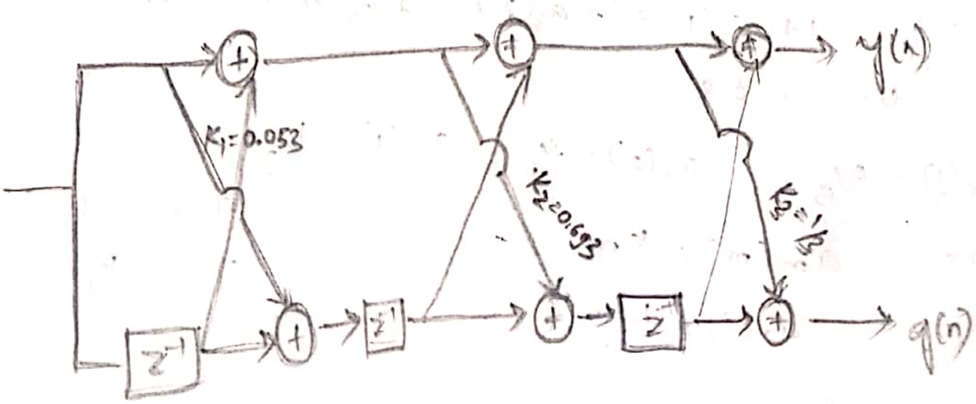
$$a_2(2) = K_2 = 0.693$$

Case(2): $m=2$, $i=1$.

$$a_1(1) = \frac{a_2(1) - a_2(2).\, a_2(1)}{1-K_2^2}$$

$$\frac{0.168 - 0.693 * 0.168}{1-(0.693)^2}$$

$$a_1(1) = 0.053 = K_1$$



(2) $H(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$ find K_1, K_2 and K_3.

Soln: $H_m(z) = 1 + \sum_{k=1}^{M} a_m(k) z^{-k}$ {here M = 3}

$$H_m(z) = 1 + a_3(1) z^{-1} + a_3(2) z^{-2} + a_3(3) z^{-3} \quad -(1)$$

$$H(z) = 1 + 13/24\, z^{-1} + 5/8\, z^{-2} + 1/3\, z^{-3} \quad -(2) \quad \{\text{given}\}$$

From (1) and (2),

$$a_3(1) = 13/24, \quad a_3(2) = 5/8, \quad a_3(3) = 1/3.$$

w.k.t, $a_m(m) = k_m$

$$\therefore a_3(3) = k_3 = 1/3.$$

Case i) $M = 3$, $i = 1$.

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1 - k_m^2}$$

$$a_2(1) = \frac{a_3(1) - a_3(3) * a_3(2)}{1 - k_3^2}.$$

$$\frac{13/24 - \{1/3 * 5/8\}}{1 - 1/9} = \frac{\left\{\frac{13-5}{24}\right\}}{8/9} = 8/24 * 9/8$$

$$a_2(1) = 9/24 = \underline{0.375}.$$

$M = 3$, $i = 2$.

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1 - k_m^2}$$

$$a_2(2) = \frac{a_3(2) - a_3(3) * a_3(1)}{1 - k_3^2}$$

$$a_2(2) = \frac{5/8 - (1/3 * 13/24)}{1 - 1/9} = 0.5$$

$$a_2(2) = k_2 = 0.5$$

Case ii) $M = 2, \; i = 1$

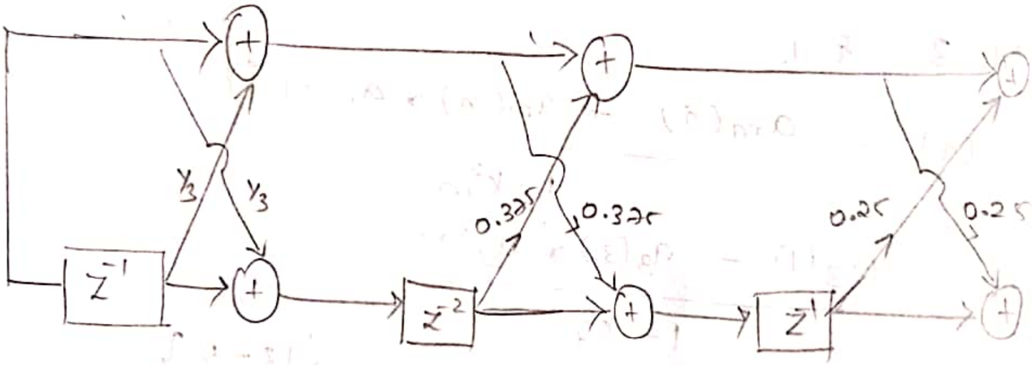
$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1 - K_m^2}$$

$$a_1(1) = \frac{a_2(1) - a_2(2) * a_2(1)}{1 - K_2^2}$$

$$= \frac{0.375 - (0.5 * 0.375)}{1 - (0.5)^2} = 0.25$$

$K_1 = \underline{a_1(1) = 0.25}$

$\therefore \; K_3 = 1/3, \quad K_2 = 0.375, \quad K_1 = 0.25.$



③. $H(z) = 1 + \frac{3}{4} z^{-1} + \frac{17}{8} z^{-2} + \frac{3}{4} z^{-3} + z^{-4}$

Soln: $H_m(z) = A_m(z) = 1 + \sum_{k=1}^{M} a_m(k) z^{-k}$

here $M = 4$,

$H_m(z) = 1 + a_4(1) + a_4(2) z^{-2} + a_4(3) z^{-3} + a_4(4) z^{-4}$ — ①

$H(z) = 1 + \frac{3}{4} z^{-1} + \frac{17}{8} z^{-2} + \frac{3}{4} z^{-3} + z^{-4}$ — ②

comparing ① and ②,

$a_4(1) = 3/4, \quad a_4(2) = 17/8, \quad a_4(3) = 3/4, \quad a_4(4) = 1$

$\therefore \; a_m(m) = K_m$

$\therefore \; a_4(4) = \underline{K_4 = 1}.$

Case i) $M=4$, $i=1$,

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1-K_m^2}$$

$$a_3(1) = \frac{a_4(1) - a_4(4) * a_4(3)}{1-K_4^2}$$

$$= \frac{\{3/4 - 3/4\}}{\{1-1\}} = 0, \quad 0/0 = \text{not defined}$$

$$\underline{a_3(1) = 0} \text{ undefined}$$

$M=4$, $i=2$,

$$a_3(2) = \frac{a_4(2) - a_4(4) * a_4(2)}{1-K_4^2} = \frac{17/8 - 17/8}{1-1} = 0$$

$$\underline{a_3(2) = 0}$$

$M=4$, $i=3$,

$$a_3(3) = \frac{a_4(3) - a_4(4)\, a_4(1)}{1-K_4^2} = \frac{3/4 - 3/4}{1-1} = 0. \text{ undefined}$$

$$a_3(3) = 0 \text{ undefined}$$

$$\therefore \underline{K_3 = 0} \text{ undefined}$$

Case ii) $M=3$, $i=1$,

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1-K_m^2}$$

$$a_2(1) = \frac{a_3(1) - a_3(3) * a_3(2)}{1-K_3^2} = \frac{0}{1} = 0 \text{ undefined}$$

$i=2$

$$a_2(2) = \frac{a_3(1) - a_3(3) * a_3(1)}{1-K_3^2} = 0$$

$$\underline{a_2(2)} = 0 \Big\} \text{ undefined}$$

$$\therefore \underline{K_2 = 0}$$

case iii) $M = 2, \quad i = 1$

$$a_{M-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-1)}{1 - K_m^2}$$

$$a_1(1) = \frac{a_2(1) - a_2(2) * a_2(1)}{1-0} = 0.$$

$$a_1(1) = K_1 = 0.$$

$\therefore \; K_4 = 1, \quad K_3 = 0, \quad K_2 = 0, \quad K_1 = 0.$

4) Given the three stage lattice filter with coefficients, $K_1 = \frac{1}{4}, \; K_2 = \frac{1}{4}, \; K_3 = \frac{1}{3}$. Determine FIR filter coefficients for the direct form structure. And draw the direct form structure. 17/10/19

Soln: $K_1 = \frac{1}{4}; \quad K_2 = \frac{1}{4}; \quad K_3 = \frac{1}{3}, \quad M = 3 \{3 \text{ stages}\}$

$$y(n) = x(n) + \sum_{k=1}^{M} a_m(k) \cdot x(n-k)$$

$$y(n) = x(n) + \sum_{k=1}^{3} a_3(k) \cdot x(n-k).$$

$$y(n) = x(n) + a_3(1) \cdot x(n-1) + a_3(2)\, x(n-2) + a_3(3) \cdot x(n-3) \quad -①.$$

$$f_0(n) = g_0(n) = x(n)$$

$$f_1(n) = x(n) + K_1 x(n-1).$$

$$g_1(n) = K_1 x(n) + x(n-1) = K_1 f_0(n) + g_0(n-1)$$

$$f_2(n) = f_1(n) + K_2\, g_1(n-1) \quad -②.$$

but, $g_1(n-1) = K_1 x(n-1) + x(n-2)$

subs $f_1(n)$ & $g_1(n-1)$ in (2)

$$f_2(n) = x(n) + K_1 x(n-1) + K_2 K_1 x(n-1) + K_2\, x(n-2)$$

$$g_2(n) = K_2 f_1(n) + g_1(n-1) \quad -⑤.$$

$$y(n) = f_3(n) = f_2(n) + K_3\, g_2(n-1) \quad -③.$$

but, $g_2(n-1) = k_2 f_1(n-1) + g_1(n-2)$

subs $f_2(n)$ & $g_3(n-1)$ in eqn ③.

$$f_3(n) = x(n) + k_1 x(n-1) + k_1 k_2 x(n-1) + k_2 x(n-2) + k_3 k_2 f_1(n-1) + k_3 g_1(n-2)$$

$$f_3(n) = x(n) + k_1(1+k_2) x(n-1) + k_2 x(n-2) + k_2 k_3 x(n-1) + k_1 k_2 k_3 x(n-2)$$
$$+ k_1 k_3 x(n-2) + k_3 x(n-3).$$

$$y(n) = f_3(n) = x(n) + (k_1 + k_1 k_2 + k_3) x(n-1) + (k_2 + k_1 k_2 k_3 + k_1 k_3) x(n-2)$$
$$+ k_3 x(n-3) \quad \text{—} ④$$

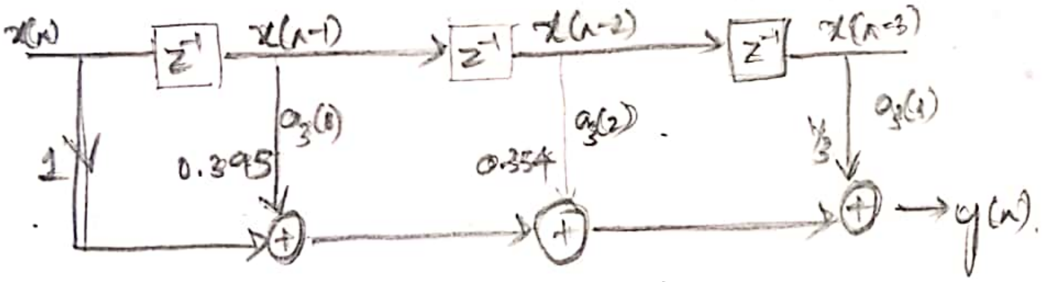
comparing ① with ④,

$a_3(1) = k_1 + k_1 k_2 + k_3 = (1/4) + (1/16) + (1/12) = 0.395.$

$a_3(2) = k_2 + k_1 k_2 k_3 + k_1 k_3 = (1/4) + (1/16 \times 1/3) + (1/12) = 0.354$

$a_3(3) = k_3 = 1/3$

Direct form structure:



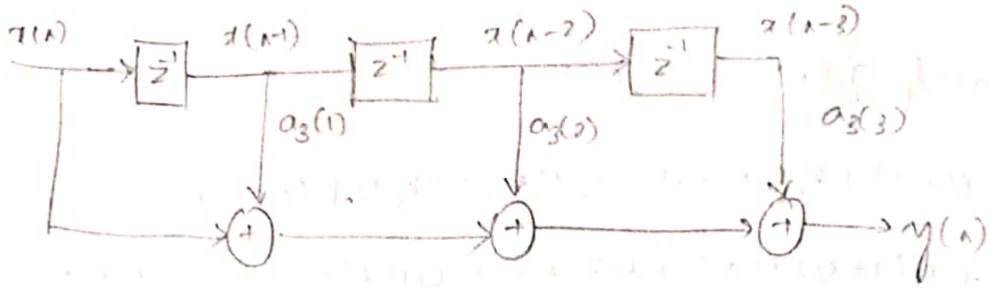
5) Let the coefficients of 3 stage FIR lattice structure be, $k_1 = 0.1$, $k_2 = 0.2$, $k_3 = 0.3$. Find the coefficients of direct form FIR filter and draw direct form structure.

Soln: Refer previous problem for solution.

$a_3(1) = k_1 + k_1 k_2 + k_3 = (0.1) + (0.1 * 0.2) + 0.3 = 0.18$

$a_3(2) = k_2 + k_1 k_2 k_3 + k_1 k_3 = (0.2) + (0.1 * 0.2 * 0.3) + (0.2 \times 0.3)$
$= 0.236$

$a_3(3) = k_3 = 0.3.$



5) Consider an FIR lattice filter with coefficients $K_1 = 0.65$, $K_2 = -0.34$, $K_3 = 0.8$. Find its impulse response and draw the equivalent direct form structure.

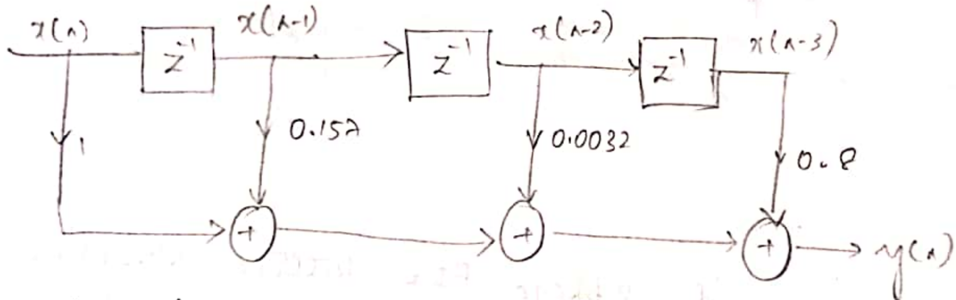
Soln: Refer problem 4;

$$a_3(1) = K_1 + K_1K_2 + K_2K_3 = (0.65) + (0.65 \times -0.34) + (-0.34 \times 0.8)$$
$$= 0.157$$

$$a_3(2) = K_1K_2K_3 + K_1K_3 + K_2 = (0.65 \times -0.34 \times 0.8) + (0.65 \times 0.8) + (-0.34$$

$$a_3(2) = 0.0032$$

$$a_3(3) = 0.8$$



To find impulse response,

from eqn ① of problem ④.

$$y(n) = x(n) + \sum_{k=1}^{M} a_m(k)\, x(n-k)$$

$$x(n) + a_3(1)x(n-1) + a_3(2)x(n-2) + a_3(3)x(n-3)$$

$$y(n) = x(n) + 0.157\, x(n-1) + 0.0032\, x(n-2) + 0.8\, x(n-3)$$

$\downarrow$ ZT,

$$Y(z) = X(z) + 0.157 z^{-1} X(z) + 0.0032 z^{-2} X(z) + 0.8 z^{-3} X(z)$$

$$Y(z) = \left[1 + 0.152z^{-1} + 0.0032z^{-2} + 0.8z^{-3}\right]X(z)$$

$$h(z) = \frac{Y(z)}{X(z)} = 1 + 0.152z^{-1} + 0.0032z^{-2} + 0.8z^{-3}$$

$\downarrow$ IZT

$$h(n) = \delta(n) + 0.152\,\delta(n-1) + 0.0032\,\delta(n-2) + 0.8\,\delta(n-3)$$

1) Given $H(z) = 1 + 2.88z^{-1} + 3.408z^{-2} + 1.74z^{-3} + 0.4z^{-4}$

i) Sketch direct form structure.

ii) Sketch lattice form structure.

iii) Find input/output equation

iv) Determine $h(n)$.

Soln: $H(z) = 1 + 2.88z^{-1} + 3.408z^{-2} + 1.74z^{-3} + 0.4z^{-4}$. — (1)

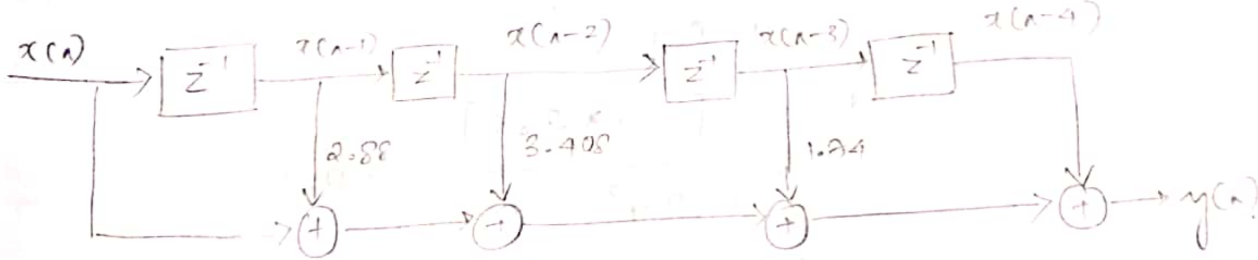
direct form:

$$\frac{Y(z)}{X(z)} = 1 + 2.88z^{-1} + 3.408z^{-2} + 1.74z^{-3} + 0.4z^{-4}$$

$$Y(z) = X(z) + 2.88z^{-1}X(z) + 3.408z^{-2}X(z) + 1.74z^{-3}X(z) + 0.4z^{-4}X(z) \quad \text{— (1)}$$

$\downarrow$ IZT.

$$y(n) = x(n) + 2.88\,x(n-1) + 3.408\,x(n-2) + 1.74\,x(n-3) + 0.42\,x(n-4).$$



ii) Lattice form structure:-

$M = 4$,
$$H_m(z) = 1 + \sum_{k=1}^{M} a_m(k)\,z^{-k}$$

$$H_4(z) = 1 + a_4(1)z^{-1} + a_4(2)z^{-2} + a_4(3)z^{-3} + a_4(4)z^{-4} \quad \text{— (2)}$$

Comparing (1) and (2),

$a_4(1) = 2.88$, $a_4(2) = 3.408$, $a_4(3) = 1.74$

$a_4(4) = 0.4$

$\therefore K_4 = a_4(4) = \underline{0.4}$

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1-k_m^2}$$

Case i) $m=4, \quad i=1$

$$a_3(1) = \frac{a_4(1) - a_4(4) * a_4(3)}{1-(0.4)^2}$$

$$= \frac{2.88 - [0.4 * 1.74]}{1-(0.4)^2} = 2.6$$

$$a_3(1) = 2.6$$

$M=4, \quad i=2$

$$a_3(2) = \frac{a_4(2) - a_4(4) * a_4(2)}{1-k_m^2}$$

$$\frac{3.408 - [0.4 * 3.408]}{1-0.4^2} = \underline{\underline{2.4342}}$$

$$a_3(2) = 2.434$$

$M=4, \quad i=3,$

$$a_3(3) = \frac{a_4(3) - a_4(4) * a_4(1)}{1-0.4^2}$$

$$= \frac{1.24 - [0.4 * 2.88]}{1-0.4^2} = 0.7$$

$$a_3(3) = 0.7 = k_3$$

Case ii) $M=3, \quad i=1,$

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) * a_m(m-i)}{1-k_m^2}$$

$$a_2(1) = \frac{a_3(1) - a_3(3) * a_3(2)}{1-k_3^2}$$

$$a_2(1) = \frac{2.6 - (0.7 * 2.434)}{1 - (0.7)^2}$$

$$a_2(1) = 1.7572$$

$i = 2$,

$$a_2(2) = \frac{a_3(2) - a_3(3) * a_3(1)}{1 - (0.7)^2}$$

$$\frac{2.434 - [0.7 * 2.6]}{1 - 0.7^2}$$

$$a_2(2) = 1.2039$$

$$\therefore \underline{K_2 = 1.203}$$

$M = 2, \; i = 1$

$$a_1(1) = \frac{a_2(1) - a_2(2) * a_2(1)}{1 - K_2^2}$$

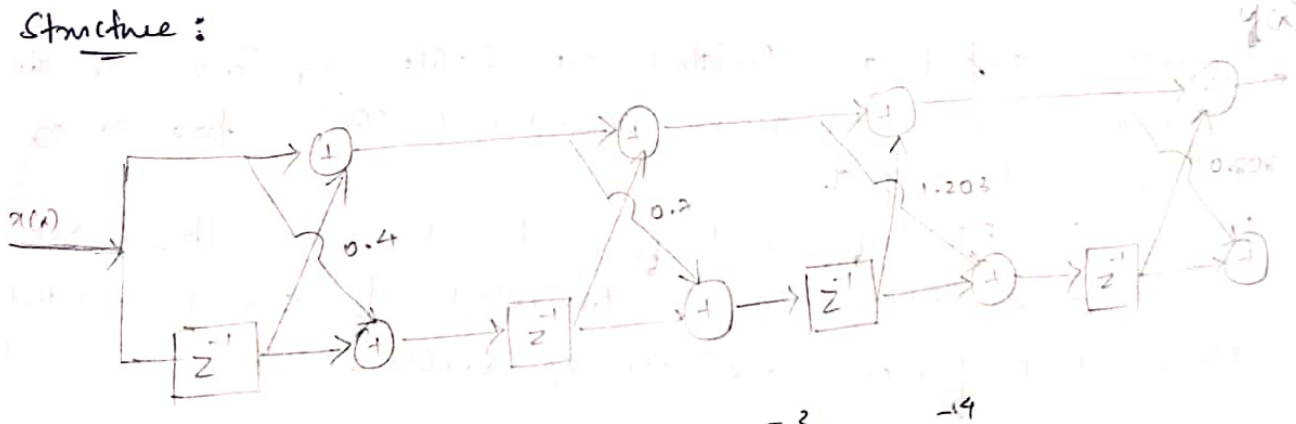
$$= \frac{1.752 - [1.203 * 1.757]}{1 - 1.203^2} = 0.808$$

$$a_1(1) = 0.806$$

$$K_1 = 0.806$$

$K_4 = 0.4, \quad K_3 = 0.7, \quad K_2 = 1.203, \quad K_1 = 0.808$

Structure :



$$H(z) = 1 + 2.88z^{-1} + 3.408z^{-2} + 1.74z^{-3} + 0.4z^{-4}$$

$$Y(z) = X(z) + 2.88z^{-1}X(z) + 3.408z^{-2}X(z) + 1.74z^{-3}X(z) + 0.4z^{-4}X(z)$$

$$y(n) = x(n) + 2.88x(n-1) + 3.408x(n-2) + 1.74x(n-3) + 0.4x(n-4)$$

$$h(n) = \delta(n) + 2.88\delta(n-1) + 3.408\delta(n-2) + 1.74\delta(n-3) + 0.4\delta(n-4)$$

→ Properties of FIR filter: 18/10/19

1. FIR filters are inherently stable.
2. FIR filters have linear phase. (Symmetric and Anti symmetric properties).
3. FIR filter needs higher orders for similar magnitude response, compared to IIR filter.

Property 1:

* Inherent stability of FIR filter:

The difference equation of FIR filter of length capital M is given by,

$$y[n] = \sum_{k=0}^{M-1} b_k x[n-k]$$

$$y[n] = b_0 x(n) + b_1 x(n-1) + \dots b_{M-1} x[n-(M-1)]$$

the coefficient b_k is related to unit sample response as,

$$h[n] = \begin{cases} b_n & 0 \le n \le M-1 \\ 0 & \text{otherwise.} \end{cases}$$

Bounded input and Bounded o/p stable systems are the systems which produces bounded output for every bounded input.

Here the FIR filter coefficient $b_0, b_1, \dots b_{M-1}$ are stable, then $y[n]$ is bounded if $x(n)$ is bounded.

Hence FIR filters are always stable.

Property 2:

Symmetric / Anti symmetric FIR filter and linear phase property:-

The symmetric / Antisymmetric of the unit sample response of FIR filter is related to their linearity of their phase.

The unit sample response of FIR filter is symmetric if $h(n) = h(N-1-n)$ and it is antisymmetric if,

$h(n) = -h(N-1-n)$.

where $n = 0 \longrightarrow N-1$.

case i). If N is symmetric and even,

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega n} + \sum_{n=N/2}^{N-1} h(n) e^{-j\omega n}.$$

2nd summation;

$$n_{new} = N-1-n$$

If $n = N/2$: $n_{new} = N-1-N/2 = N/2 - 1$.

If $n = N-1$, $n_{new} = 0$.

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega n} + \sum_{n=N/2-1}^{0} h(N-1-n) e^{-j\omega(N-1-n)}$$

$$h(N-1-n) = h(n) \quad \{\text{symmetric}\}$$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega n} \cdot 1 + \sum_{n=0}^{N/2-1} h(n) e^{-j\omega(N-1)} \cdot e^{j\omega n}$$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega n} e^{-j\omega\frac{(N-1)}{2}} \cdot e^{j\omega\frac{(N-1)}{2}} + \sum_{n=0}^{N/2-1} h(n) e^{-j\omega\frac{(N-1)}{2}} \cdot e^{-j\omega\frac{(N-1)}{2}} e^{+j\omega n}.$$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega\frac{(N-1)}{2}} \left[e^{-j\omega n} e^{j\omega\frac{(N-1)}{2}} + e^{-j\omega\frac{(N-1)}{2}} \cdot e^{j\omega n} \right]$$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega\frac{(N-1)}{2}} \left[e^{j\omega\left(\frac{N-1}{2} - n\right]} + e^{-j\omega\left[\frac{N-1}{2} - n\right]} \right]$$

$$\theta = \omega\left[\frac{N-1}{2} - n\right]$$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n) e^{-j\omega\frac{(N-1)}{2}} \left[2\cos\omega\left(\frac{N-1}{2} - n\right)\right]$$

the polar general form,

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

$$|H(\omega)| = 2\sum_{n=0}^{N/2-1} h(n)\cos\omega\left(\frac{N-1}{2}-n\right).$$

$$\underline{|H(\omega)} = \begin{cases} -\omega\left(\frac{N-1}{2}\right) & \text{for } |H(\omega)| > 0 \\ -\omega\left(\frac{N-1}{2}\right)+\pi & \text{for } |H(\omega)| < 0. \end{cases}$$

Since $\left(\frac{N-1}{2}\right)$ is constant, phase varies linearly with ω.

<u>case (ii) when N is symmetric and odd</u> :- 24/10/19

The fourier transform of unit impulse response is given by,

$$H(\omega) = \sum_{n=0}^{N-1} h(n) e^{-j\omega n}$$

$$H(\omega) = \sum_{n=0}^{\frac{N-3}{2}} h(n)e^{-j\omega n} + h\left(\frac{N-1}{2}\right)e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N-1}{2}}^{N-1} h(n)e^{-j\omega n}$$

2nd summation,

Let $n_{new} = N-1-n$.

If $n = \frac{N+1}{2}$, $n_{new} = N-1-\frac{N}{2}-\frac{1}{2} = \frac{N}{2}-\frac{3}{2}$

$= \frac{N-3}{2}.$

If $n = N-1$, $n_{new} = N-1-N+1$

$= 0.$

$$H(\omega) = \sum_{n=0}^{\frac{N-3}{2}} h(n)e^{-j\omega n} + h\left(\frac{N-1}{2}\right)e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N-3}{2}}^{0} h(N-1-n)e^{-j\omega(N-1-n)}$$

since symmetric, $h(N-1-n) = h(n)$.

$$H(\omega) = h\left(\frac{N-1}{2}\right)e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n)e^{-j\omega n} + \sum_{n=0}^{\frac{N-3}{2}} h(n)\, e^{-j\omega(N-1)} e^{j\omega n}$$

$$H(\omega) = h\left(\frac{N-1}{2}\right)e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n)\left[e^{-j\omega n}\cdot 1 + e^{-j\omega(N-1)}e^{j\omega n}\right]$$

$$H(\omega) = h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{N-3/2} h(n) \left[e^{-j\omega n} \cdot e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{j\omega\left(\frac{N-1}{2}\right)} + e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{j\omega n} \right]$$

$$H(\omega) = h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega\left(\frac{N-1}{2}\right)} \left[e^{j\omega\left(\frac{N-1}{2}-n\right)} + e^{-j\omega\left(\frac{N-1}{2}-n\right)} \right]$$

here $\theta = \omega\left(\frac{N-1}{2} - n\right)$

$$H(\omega) = h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega\left(\frac{N-1}{2}\right)} \left[2\cos\omega\left(\frac{N-1}{2} - n\right) \right]$$

$$H(\omega) = \left[h\left(\frac{N-1}{2}\right) + 2\sum_{n=0}^{\frac{N-3}{2}} h(n) \cos\omega\left(\frac{N-1}{2} - n\right) \right] e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$$

$$|H(\omega)| = h\left(\frac{N-1}{2}\right) + 2\sum_{n=0}^{\frac{N-3}{2}} h(n) \cos\omega\left(\frac{N-1}{2} - n\right)$$

$$\angle H(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right), & |H(\omega)| > 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi, & |H(\omega)| < 0 \end{cases}$$

Since $\left(\frac{N-1}{2}\right)$ is constant, phase of $H(\omega)$ varies linearly w.r.t ω, $\therefore$ FIR filter are linear phase filters.

case (ii) where N is even and Antisymmetric.

The fourier transform is

$$H(\omega) = \sum_{n=0}^{N-1} h(n) e^{-j\omega n}$$

$$H(\omega) = \sum_{n=0}^{\frac{N}{2}-1} h(n) e^{-j\omega n} + \sum_{n=N/2}^{N-1} h(n) e^{-j\omega n}$$

2nd summation:

$$n_{new} = N-1-n$$

$n_{new} = N-1-N/2 \Rightarrow N/2 - 1$

$n_{new} = N-1-N+1 \Rightarrow 0.$

$$H(\omega) = \sum_{n=0}^{N/2-1} h(n)e^{-j\omega n} + \sum_{n=N/2-1}^{0} h(N-1-n)\, e^{-j\omega(N-1-n)}$$

$$h(N-1-n) = -h(n)$$

$$= \sum_{n=0}^{N/2-1} h(n)e^{-j\omega n} - \sum_{n=0}^{N/2-1} h(n)\, e^{-j\omega(N-1)} \cdot e^{j\omega n}$$

$$\sum_{n=0}^{N/2-1} h(n)e^{-j\omega n} \cdot 1 - \sum_{n=0}^{N/2-1} h(n)e^{-j\omega(N-1)} \cdot e^{j\omega n}$$

$$\sum_{n=0}^{N/2-1} h(n)e^{-j\omega n} \cdot e^{-j\omega\frac{(N-1)}{2}} \cdot e^{j\omega\left(\frac{N-1}{2}\right)} - \sum_{n=0}^{N/2-1} h(n)e^{-j\omega\frac{(N-1)}{2}} \cdot e^{-j\omega\frac{(N-1)}{2}}$$

$$\sum_{n=0}^{N/2-1} h(n)e^{-j\omega\frac{(N-1)}{2}} \left[e^{-j\omega n}\, e^{j\omega\left(\frac{N-1}{2}\right)} - e^{-j\omega\left(\frac{N-1}{2}\right)} e^{j\omega n} \right]$$

$$\sum_{n=0}^{N/2-1} h(n)e^{-j\omega\left(\frac{N-1}{2}\right)} \left[e^{j\omega\left(\frac{N-1}{2}-n\right)} - e^{-j\omega\left(\frac{N-1}{2}-n\right)} \right]$$

$$\theta = \omega\left(\frac{N-1}{2} - n\right)$$

$$\sum_{n=0}^{N/2-1} h(n)e^{-j\omega\left(\frac{N-1}{2}\right)} \left[2j \sin\omega\left(\frac{N-1}{2} - n\right) \right]$$

$$\cancel{H(\omega) = e^{-j\omega\left(\frac{N-1}{2}\right)}} \left[h(n) \sin \right.$$

$$H(\omega) = 2j \sum_{n=0}^{N/2-1} \left[h(n) \sin\omega\left(\frac{N-1}{2} - n\right) \right] e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$|H(\omega)| = 2j \sum_{n=0}^{N/2-1} \left[h(n) \sin\omega\left(\frac{N-1}{2} - n\right) \right.$$

$$\angle H(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right), & |H(\omega)| > 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi/2 & |H(\omega)| < 0. \end{cases}$$

case iv) If N is odd and Antisymmetric

$$H(\omega) \sum_{n=0}^{N-1} h(n) e^{-j\omega n}$$

$$H(\omega) = \sum_{n=0}^{N-3/2} h(n) e^{-j\omega n} + h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N-1}{2}}^{N-1} h(n) e^{-j\omega n}$$

2nd summation:

$n_{new} = N-1-n.$

$$n = \frac{N-1}{2} \Rightarrow n_{new} = N-1-\frac{N}{2}+\frac{1}{2} = N/2 - 3/2 \Rightarrow \frac{N-3}{2}$$

$$n = N-1 \quad n_{new} = N-1-N+1 = 0.$$

$$H(\omega) = \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega n} + h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N-3}{2}}^{0} h(N-1-n) e^{-j\omega(N-1-n)}$$

$$h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega n} - \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega(N-1)} \cdot e^{j\omega n}.$$

$$h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega n} \cdot e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{+j\omega\left(\frac{N-1}{2}\right)}$$

$$- \sum_{n=0}^{N-3/2} h(n) e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{-j\omega\left(\frac{N-1}{2}\right)} \cdot e^{j\omega n}$$

$$h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) e^{-j\omega\left(\frac{N-1}{2}\right)} \left\{ e^{-j\omega n} \cdot e^{j\omega\left(\frac{N-1}{2}\right)} - e^{j\omega n} \cdot e^{-j\omega\left(\frac{N-1}{2}\right)} \right\}$$

$$h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{N-3/2} h(n) e^{-j\omega\left(\frac{N-1}{2}\right)} \left\{ e^{j\omega\left(\frac{N-1}{2}-n\right)} - e^{-j\omega\left(\frac{N-1}{2}-n\right)} \right\}$$

$$h\left(\frac{N-1}{2}\right) \underline{e}^{-j\omega\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{N-3}{2}} h(n) \underline{e}^{-j\omega\left(\frac{N-1}{2}\right)} \left\{ 2j \sin \omega\left(\frac{N-1}{2}-n\right) \right\}$$

$$\therefore H(\omega) = \left\{ h\left(\frac{N-1}{2}\right) + 2j \sum_{n=0}^{\frac{N-3}{2}} e^{-j\omega\left(\frac{N-1}{2}\right)} \right\} e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$\angle H(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right) & |H(\omega)| > 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi/2 & |H(\omega)| < 0 \end{cases}$$

$$|H(\omega)| = h\left(\frac{N-1}{2}\right) + 2j \sum_{n=0}^{\frac{N-3}{2}} e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$\therefore \left(\frac{N-1}{2}\right)$ is constant and hence it is linear.

Property 3: Magnitude Characteristics and Order of FIR filter.

The magnitude response is expressed as,

$$1-\delta_1 \le |H(\omega)| \le 1+\delta_1, \quad \text{for } 0 \le \omega \le \omega_p$$
$$0 \le |H(\omega)| \le \delta_2 \quad \text{for } \omega_s \le \omega \le \pi$$

∴ The order N is given as

$$N = \frac{-10\log_{10}(\delta_1\delta_2) - 15}{14\,\Delta f}$$

where $\Delta f = \frac{\omega_s - \omega_p}{2\pi}$ is the transmission band

and $\omega_s = 2\pi f_s$; $\omega_p = 2\pi f_p$.

∴ $\Delta f = f_s - f_p$

The length of the filter is $\boxed{M = N}$

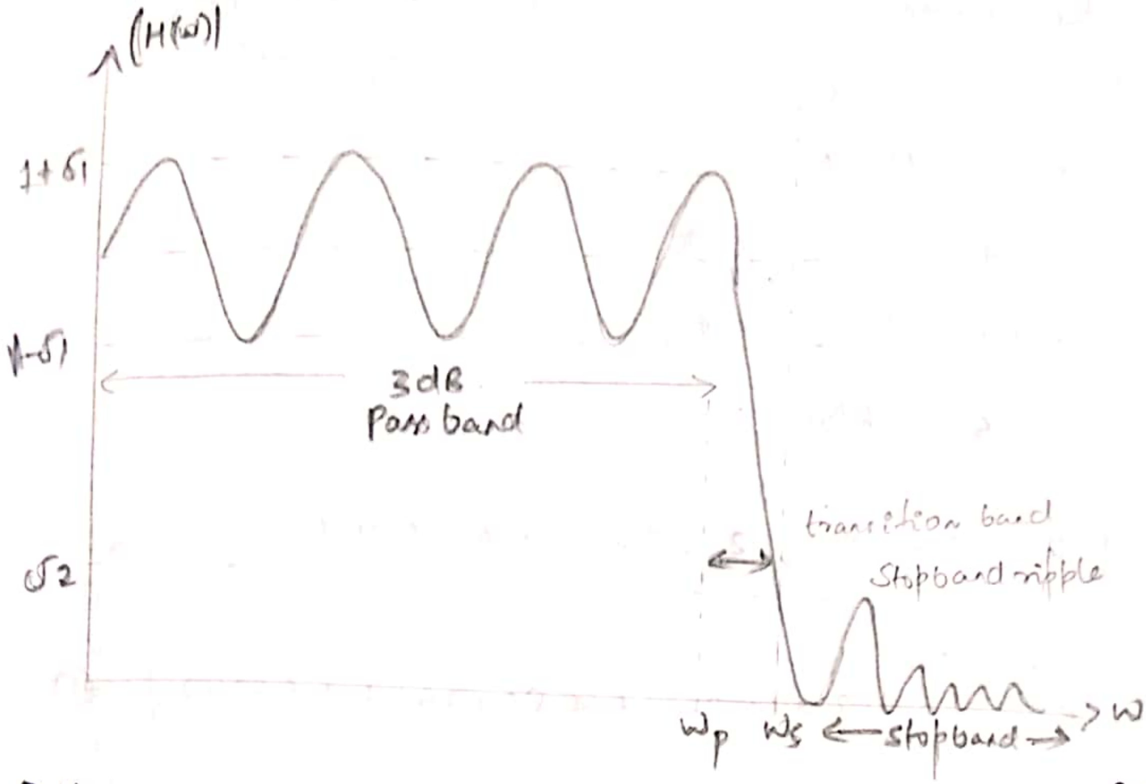
or $M = N = k\left[\frac{2\pi}{\omega_2 - \omega_1}\right]$

width of main lobe = $k\left[\frac{2\pi}{M}\right]$

For similar magnitude specifications of FIR have higher order than IIR.

∴ FIR do not have feedback, hence they need long sequence of h(n) to get sharp cut off filters.

∴ Because of increased number of co-efficients, FIR requires large processing time. this processing time can be reduced using FFT Algorithm..



Problems: 25/10/19

The filter response of linear phase FIR filter is,

$$H(e^{j\omega}) = e^{-j3\omega}\left[2 + 1.8\cos 3\omega + 1.2\cos 2\omega + 0.5\cos\omega\right].$$

Find the impulse response sequence of the filter.

Soln: $H(e^{j\omega}) = e^{-j3\omega}\left[2 + 1.8\cos 3\omega + 1.2\cos 2\omega + 0.5\cos\omega\right]$ — (*).

$$e^{-j3\omega} = e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$\frac{N-1}{2} = 3 \Rightarrow \underline{\underline{N=7}}$$

∴ N is odd and symmetric {since we have only 'cos' terms}.

w.k.t,

$$H(\omega) = \left[2\sum_{n=0}^{\frac{N-3}{2}} h(n)\cos\omega\left(\frac{N-1}{2}-n\right) + h\left(\frac{N-1}{2}\right)\right]e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$2\sum_{n=0}^{2} h(n)\cos\omega(3-n) + h(3) \qquad \{\because \text{substituting for } N=7\}$$

$$2h(0)\cos 3\omega + 2h(1)\cos 2\omega + 2h(2)\cos\omega + h(3) \text{ — (1).}$$

$h(3) = 2$

$2h(0) = 1.8, \quad h(0) = 0.9$

$2h(1) = 1.2; \quad h(1) = 0.6$

$2h(2) = 0.5; \quad h(2) = 0.25$

{comparing (1) and (*)

Since the order is 7, we have found only 4 coefficients, to find the next 3 coefficients,

$$h(n) = h(N-1-n)$$

$h(4) = h(7-1-4) = h(2) = 0.25$

$h(5) = h(7-1-5) = h(1) = 0.6$

$h(6) = h(7-1-6) = h(0) = 0.9$

$\therefore h(n) = \{0.9, 0.6, 0.25, 2, 0.25, 0.6, 0.9\}.$

2) $H(\omega) = e^{-j3\omega}[5 + 3.2\cos 3\omega + 4.8\cos 2\omega + 1.6\cos\omega]$ — ①

Soln: $e^{-j\omega\left(\frac{N-1}{2}\right)} = e^{-j3\omega}$

$\frac{N-1}{2} = 3$; $\underline{N=7}$

N is odd and symmetric,

$$H(\omega) = \left[2\sum_{n=0}^{\frac{N-3}{2}} h(n)\cos\omega\left(\frac{N-1}{2} - n\right) + h\left(\frac{N-1}{2}\right)\right]e^{-j\omega\left(\frac{N-1}{2}\right)}$$

Substitute $N=7$

$$= 2\sum_{n=0}^{2} h(n)\cos\omega(3-n) + h(3)$$

$2h(0)\cos\omega(3) + 2h(1)\cos 2\omega + 2h(2)\cos\omega + h(3)$ — ②

$\therefore$ Comparing ② and ①,

$h(3) = 5$

$2h(2) = 1.6$; $h(2) = 0.8$

$2h(1) = 4.8$; $h(1) = 2.4$

$2h(0) = 3.2$; $h(0) = 1.6$

$\therefore h(n) = h(N-1-n)$

$h(4) = h(7-1-4) = h(2) = 1.6$

$h(5) = h(1) = 2.4$

$h(6) = h(0) = 0.8$

$h(n) = \{0.8, 2.4, 0.8, 5, 0.8, 2.4, 0.8\}$.

→ Different types of windows:

1. Rectangular window
2. Bartlett window or triangular window
3. Blackman window
4. Hamming window
5. Hanning window.
6. Kaiser window.

→ RECTANGULAR WINDOW :-

The rectangular window of length M is given by,

$$w_R(n) = \begin{cases} 1 & \text{for } n = 0 \ldots M-1 \\ 0 & \text{otherwise} \end{cases}$$

Fourier transform of rectangular window,

$$W_R(\omega) = \sum_{n=0}^{M-1} w_R(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{M-1} (e^{-j\omega})^n$$

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$$

$$W_R(\omega) = \frac{1-e^{-j\omega M}}{1-e^{-j\omega}}$$

$$\frac{e^{-j\omega\frac{M}{2}} \cdot e^{+j\omega\frac{M}{2}} - e^{-j\omega\frac{M}{2}} \cdot e^{-j\omega\frac{M}{2}}}{e^{-j\frac{\omega}{2}} \cdot e^{+j\frac{\omega}{2}} - e^{-j\omega/2} \cdot e^{-j\omega/2}}$$

$$\frac{e^{-j\frac{wM}{2}}\left[e^{\frac{jwM}{2}} - e^{\frac{-jwn}{2}}\right]}{e^{-\frac{jw}{2}}\left[e^{\frac{jw}{2}} - e^{-\frac{jw}{2}}\right]}$$

$$W_R(w) = \frac{e^{-\frac{jwM}{2}} \cdot e^{\frac{jw}{2}}\left[2j\sin\left(\frac{wM}{2}\right)\right]}{2j\sin\left(\frac{w}{2}\right)}$$

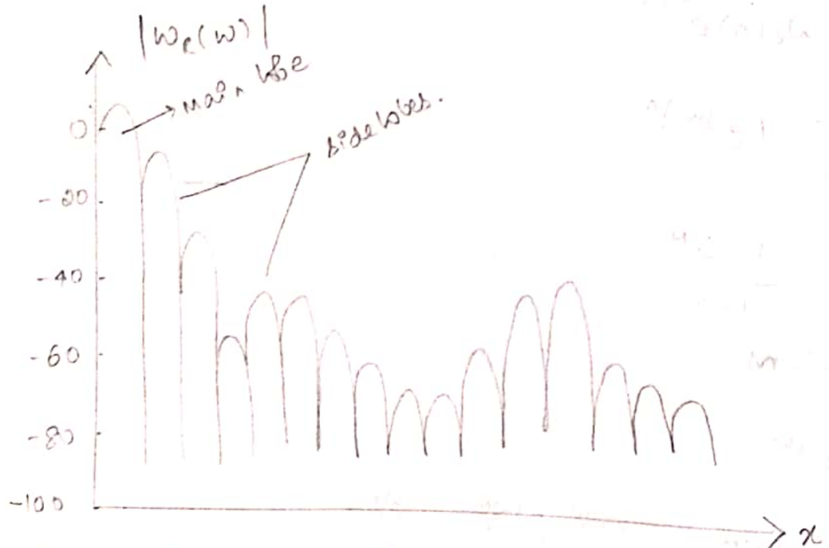
$$\frac{e^{-j\frac{w}{2}(M-1)}\sin\left(\frac{wM}{2}\right)}{\sin\left(w/2\right)}$$

$$W_R(w) = |W_R(w)|\, e^{j\angle W_R(w)}$$

$$|W_R(w)| = \frac{\sin\left(\frac{wM}{2}\right)}{\sin w/2}$$ is the magnitude response of Rectangular window.

$$\angle W_R(w) = \frac{w}{2}(M-1)$$

If $M = 50$, the response of rectarangular window is as shown below.



* From the magnitude spectrum it is clear that there is one main lobe followed by side lobes.
* As M increases the main lobe becomes narrower. The area under the side lobes remains ~~same~~ same irrespective of change in 'M'.

→ The unit sample response of the FIR filter is

$$h(n) = h_d(n)\, w_R(n) \quad \text{--- (4)}.$$

Sub (1) in (4)

$$h(n) = \begin{cases} h_d(n) & \text{for } n = 0 \ldots M-1 \\ 0 & \text{otherwise.} \end{cases}$$

Taking Fourier transform to (4).

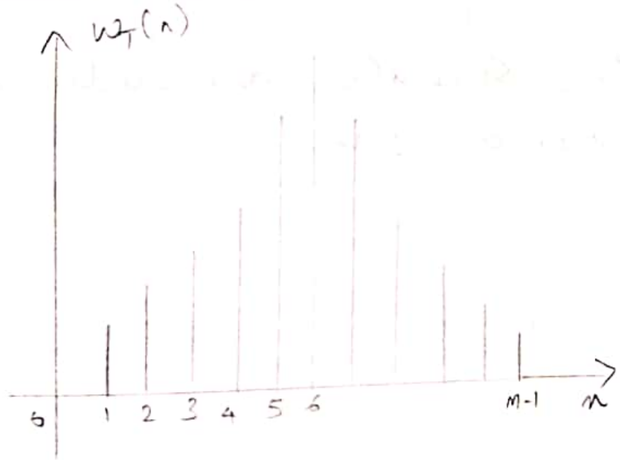
$$H(\omega) = H_d(\omega) \cdot W_R(\omega).$$

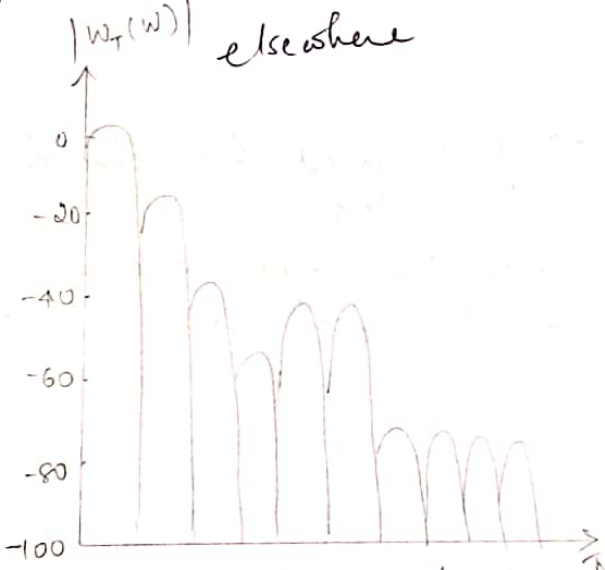
2) Bartlett or Triangular window

Barlett window is mathematically expressed as,

$$w_T(n) = \begin{cases} 1 - \dfrac{2\left| n - \frac{M-1}{2} \right|}{M-1} & \text{for } n = 0, 1 \ldots M-1 \\ 0 & \text{elsewhere} \end{cases}$$

Time domain sketch:





Magnitude response

3) Hamming window :-

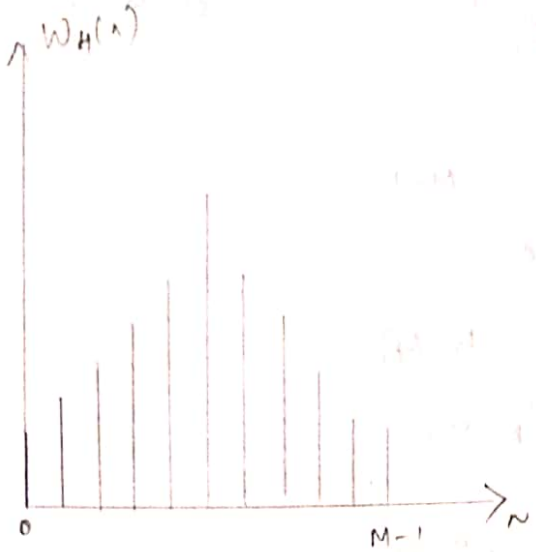
Hamming window is the most commonly used in speech processing and it is given as,

$$w_H(n) = \begin{cases} 0.54 - 0.46 \cos \dfrac{2\pi n}{M-1}, & \text{for } n = 0, 1, \ldots M-1 \\ 0 & \text{otherwise} \end{cases}$$

→ It is used for spectrum analysis speech and music processing.

→ It has narrow main lobe, sidelobes are reduced

→ The first and last samples are ~~zero~~ not zero.



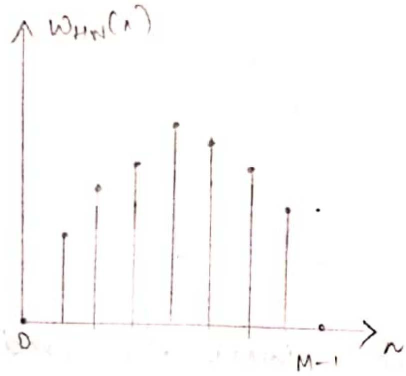
4) Hanning window :-

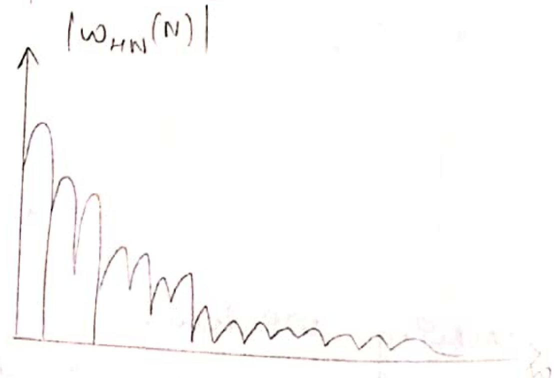
$$W_{HN} = \begin{cases} \frac{1}{2}\left[1 - \cos\left(\frac{2\pi n}{M-1}\right)\right] & \text{for } n = 0 \ldots M-1 \\ 0 & \text{otherwise.} \end{cases}$$

→ It is used for spectrum analysis speech and music processing.

→ It has narrow main lobe, sidelobes are reduced

→ The first and last samples are zero.





Design the symmetric FIR Low-pass filter whose desired frequency response is given by, 26/10/19

$$H_d(\omega) = \begin{cases} e^{-j\alpha\omega} & |\omega| \le \omega_c \\ 0 & \text{otherwise.} \end{cases}$$

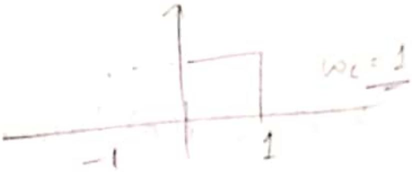
The length of the filter should be 7. The value of ω_c should be ~~rad/se~~ 1 radian/sample. Use rectangular window.

%

desired frequency response → $H_d(\omega) = \begin{cases} e^{-j\alpha\omega} & |\omega| \le \omega_c \\ 0 & \text{otherwise} \end{cases}$ $(-\pi \le |\omega| \le \pi)$.

* desired impulse response → $h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega.$

* $h(n) = h_d(n)\,w(n).$



Soln: $h(n) = h_d(n)\,w_R(n)$

$$h_d(n) = \frac{1}{2\pi}\int_{\pi}^{\pi} H_d(\omega)e^{j\omega n}.\,d\omega$$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)e^{j\omega n}.\,d\omega$$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\alpha\omega}.\;e^{j\omega n}\,d\omega$$

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)}.\,d\omega \quad \text{--- (1)}.$$

Case i) $n \neq \alpha$,

$$h_d(n) = \frac{1}{2\pi}\left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\right]_{-\omega_c}^{\omega_c}$$

$$\frac{1}{2j\pi(n-\alpha)}\left[e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)}\right]$$

$$h_d(n) = \frac{2j\sin\omega_c(n-\alpha)}{2j\pi(n-\alpha)}$$

$$= \frac{\sin\omega_c(n-\alpha)}{\pi(n-\alpha)}.$$

Case ii) $n = \alpha$ in (1).

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{0}.\,d\omega = \frac{1}{2\pi}\left[\omega\right]_{-\omega_c}^{\omega_c} = \frac{2\omega_c}{2\pi} = \frac{\omega_c}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin\omega_c(n-\alpha)}{\pi(n-\alpha)}, & n \neq \alpha \\[2ex] \dfrac{\omega_c}{\pi}, & n = \alpha. \end{cases}$$

[illegible],

since $\omega_c = 1$, $\quad h_d(n) = \begin{cases} \dfrac{\sin(n-\alpha)}{\pi(n-\alpha)} & ; n \neq \alpha \\ \dfrac{1}{\pi} & n = \alpha . \end{cases}$

$h(n) = h_d(n) \cdot \omega_R(n).$

$$\omega_R(n) = \begin{cases} 1 & n = 0 \ldots M-1 \\ 0 & \end{cases}$$

$h(n) = h_d(n).$

$$h(n) = \begin{cases} \dfrac{\sin(n-\alpha)}{\pi(n-\alpha)} & n = 0 \ldots M-1 \\ 0 & \text{otherwise.} \end{cases}$$

②. A low pass filter is to be designed with the following frequency response.

$$H_d(\omega) = \begin{cases} e^{-j2\omega} & \omega \leq \pi/4 \\ 0 & \pi/4 < \omega < \pi \end{cases}$$

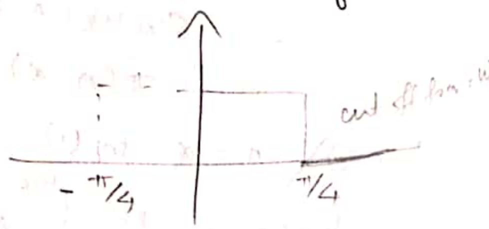
$\frac{N-1}{2} = \alpha \quad N-1 = 4 \quad \boxed{N = 5}$

determine the filter coefficient $h_d(n)$ and impulse response $h(n)$. If ω_n is rectangular window defined as

$$\omega_R(n) = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \end{cases}$$

Find the frequency response of the FIR filter.

Soln: $h_d(n) = \dfrac{1}{2\pi}\displaystyle\int_{-\omega_c}^{\omega_c} H_d(\omega)\, e^{j\omega n} \cdot d\omega$



$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\alpha\omega} \cdot e^{j\omega n} \cdot d\omega \quad = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} \cdot d\omega \quad —①$$

Case i) $(n-\alpha) \neq 0$.

$$\left.\frac{e^{j\omega(n-\alpha)}}{2\pi j(n-\alpha)}\right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi j(n-\alpha)}\left[e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)}\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[\sin[\omega_c(n-\alpha)]\,2j\right]$$

$$h_d(n) = \frac{\sin \omega_c(n-\alpha)}{\pi(n-\alpha)}$$

case ii) $n-\alpha = 0$. in ①,

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{0}\,.d\omega = \frac{\omega_c+\omega_c}{2\pi} = \frac{2\omega_c}{2\pi} = \frac{\omega_c}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin \omega_c(n-\alpha)}{\pi(n-\alpha)}, & (n-\alpha) \neq 0 \\[2ex] \dfrac{\omega_c}{\pi}, & (n-\alpha) = 0 \end{cases}$$

as, $\omega_c = \pi/4$, $\alpha = 2$

$$h_d(n) = \begin{cases} \dfrac{\sin \pi/4\,(n-2)}{\pi(n-2)}, & (n-\alpha) \neq 0 \\[2ex] \cancel{\pi/4\pi}\ \ 1/4 & (n-\alpha) = 0 \end{cases}$$

n	$h_d(n)$	$\omega_R(n)$	$h(n) = h_d(n)\,\omega_R(n)$
0	0.159	1	0.159
1	0.225	1	0.225
2	0.25	1	0.25
3	0.225	1	0.225
4	0.159	1	0.159

N ~~N~~ = 5 → odd and Symmetric.

$h(n) = \{0.159, 0.225, 0.25, 0.225, 0.159\}$

N is odd, $\therefore$ odd and symmetric.

frequency response

$$\therefore H(\omega) = \left[h\left(\frac{N-1}{2}\right) + 2 \sum_{n=0}^{\frac{N-3}{2}} h(n) \cos\omega\left(\frac{N-1}{2} - n\right) \right] e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$H(\omega) = \left[h(2) + 2 \sum_{n=0}^{1} h(n) \cos\omega(2-n) \right] e^{-j2\omega}$$

$$H(\omega) = h(2) + \left[2h(0)\cos 2\omega + 2h(1)\cos\omega\right] e^{-j2\omega}$$

$$= 0.25 + \left[(2 \times 0.159 \cos 2\omega) + (2 \times 0.225 \cos\omega)\right] e^{-j2\omega}$$

$$H(\omega) \quad 0.25 + \left[0.318 \cos 2\omega + 0.45 \cos\omega\right] e^{-j2\omega}$$

3). A low pass filter is to be designed with the desired frequency response

$$H_d(\omega) = \begin{cases} e^{-j3\omega} & |\omega| < \frac{3\pi}{4} \\ 0 & \frac{3\pi}{4} < \omega < \pi \end{cases}$$

Determine the filter coefficient $h_d(n)$, impulse response $h(n)$ and frequency response of the resulting filter. Use rectangular window.

Soln:

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H_d(\omega) e^{j\omega n} d\omega$$

$$\frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j3\omega} e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{+j\omega(n-3)} d\omega$$

$n-3 \neq 0$ $n \neq 3$.

$$\cancel{\frac{1}{2\pi} \left[\frac{e^{-j3\omega}}{-j3}\right]_{-\omega_c}^{\omega_c}} \quad \frac{1}{2\pi} \left[\frac{e^{j\omega(n-3)}}{j(n-3)}\right]_{-\omega_c}^{\omega_c}$$

$$\cancel{\frac{1}{3j \times 2\pi}} \Big[$$

$$\frac{1}{2\pi \times j(n-3)}\left[e^{jw_c(n-3)} - e^{+jw_c(n-3)}\right]$$

$$\frac{1}{2\pi \times j(n-3)}\left[2j \sin w_c(n-3)\right]$$

$$h_d(n) = \frac{\sin w_c(n-3)}{\pi(n-3)}$$

ii). $n-3=0 \Rightarrow n=3.$

$$\frac{1}{2\pi}\int_{-w_c}^{w_c} e^{jw(0)}.\,dw = \frac{1}{2\pi}\left[w_c + w_c\right] = \frac{w_c}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin w_c(n-3)}{\pi(n-3)}, & n \neq 3. \\ \dfrac{w_c}{\pi}, & n = 3. \end{cases}$$

here $w_c = 3\pi/4$

$$h_d(n) = \begin{cases} \dfrac{\sin \frac{3\pi}{4}(n-3)}{\pi(n-3)}, & n \neq 3 \\ \dfrac{3}{4}, & n = 3 \end{cases}$$

to find N, $\quad e^{-j3w} = e^{-jw\left(\frac{N-1}{2}\right)}$

$$3 = \frac{N-1}{2} \Rightarrow N-1 = 6$$

$$\therefore \boxed{N = 7}$$

n	$h_d(n)$	$w_R(n)$	$h(n) = h_d(n).\,w_R(n)$
0	0.0750	1	0.0750
1	-0.159	1	-0.159
2	0.2250	1	0.2250
3	3/4	1	3/4
4	0.2250	1	0.2250
5	-0.159	1	-0.159
6	0.0750	1	0.0750

N = 7 and symmetric

$\Rightarrow$ odd and symmetric

$\therefore$ frequency response $H(\omega) = \left[h\left(\frac{N-1}{2}\right) + 2 \sum_{n=0}^{\frac{N-3}{2}} h(n) \cos\omega\left(\frac{N-1}{2} - n\right) \right] e^{\dots}$

$$H(\omega) = \left[h(3) + 2 \sum_{n=0}^{2} h(n) \cos\omega(3-n) \right] e^{-j3\omega}.$$

$$= h(3) + \{ 2h(0)\cos 3\omega + 2h(1)\cos 2\omega + 2h(2)\cos\omega \} e^{-j3\omega}.$$

$$h(n) = \{ 0.075, -0.159, 0.225, 3/4, 0.225, -0.159, 0.075 \}$$

$$\frac{3}{4} + \{ 2\times 0.075 \cos 3\omega + 2\times(-0.159)\cos 2\omega + 2\times 0.225\cos\omega \} e^{-j3\omega}$$

$$\frac{3}{4} + \{ 0.15\cos 3\omega - 0.318\cos 2\omega + 0.45\cos\omega \} e^{-j3\omega}.$$

— find $h_d(n)$, $H_\omega(n)$,

i). Same Question for Hamming window.

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H_d(\omega) e^{j\omega n} \, d\omega.$$

$$\frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j3\omega} \cdot e^{j\omega n} \cdot d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-3)} \cdot d\omega$$

$$\frac{1}{2\pi} \left[\frac{e^{j\omega(n-3)}}{j(n-3)} \right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi j(n-3)} \left[e^{j\omega_c(n-3)} - e^{-j\omega_c(n-3)} \right]$$

$$\frac{1}{2\pi j(n-3)} \left[2j \sin\omega_c(n-3) \right] = \frac{\sin\omega_c(n-3)}{\pi(n-3)}$$

Case ii) $n-3=0$,

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^0 \, d\omega = \frac{2\omega_c}{2\pi} = \frac{\omega_c}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin \omega_c(n-3)}{\pi(n-3)} & n-3 \neq 0 \\ \dfrac{\omega_c}{\pi} & n-3=0. \end{cases}$$

here $\omega_c = 3\pi/4$.

$$h_d(n) = \begin{cases} \dfrac{\sin \frac{3\pi}{4}(n-3)}{\pi(n-3)} & n \neq 3 \\ 3/4 & n=3 \end{cases}$$

Since it is Hamming window,

$$\omega_H(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right)$$

here $M=7$ {in previous problem}

n	$h_d(n)$	$\omega_H(n)$	$h(n) = h_d(n)\,\omega_H(n)$
0	0.0750	0.08	6×10^{-3}
1	−0.159	0.31	~~−0.16069~~ 0.04929
2	0.2250	0.77	~~0.0442~~ 0.17325
3	3/4	1	0.75
4	0.2250	0.77	0.17325
5	−0.159	0.31	−0.04929
6	0.0750	0.08	6×10^{-3}

$h(n) = \{6\times10^{-3}, 0.04929, 0.17325, 0.75, 0.17325, -0.0492, 6\times10^{-3}\}$.

$N=7$ i. odd and symmetric.

Frequency response for odd and symmetric sequence,

$$H(\omega) = \left[h\left(\frac{N-1}{2}\right) + 2\sum_{n=0}^{\frac{N-3}{2}} h(n)\cos\omega\left[\frac{N-1}{2} - n\right]\right] e^{-j\omega\left(\frac{N-1}{2}\right)}$$

$$H(\omega) = \left[h(3) + \left[2\sum_{n=0}^{2} h(n)\cos\omega[3-n]\right]\right] e^{-j\omega 3}.$$

$$H(\omega) = h(3) + \left\{2h(0)\cos 3\omega + 2h(1)\cos 2\omega + 2h(2)\cos\omega\right\} e^{-j3\omega}$$

$$= 0.25 + \left\{(2\times 6\times 10^{-3})\cos 3\omega + (2\times 0.0492)\cos 2\omega + (2\times 0.173)\cos\omega\right\} e^{-j3\omega}$$

$$0.25 + \left[0.012\cos 3\omega + 0.0984\cos 2\omega + 0.346\cos\omega\right] e^{-j3\omega}$$

5) Design FIR filter using Hanning window.

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & \text{for } |\omega| \le \omega_c \\ 0 & \text{otherwise.} \end{cases}$$

The length of filter should be 7 and $\omega_c = 1\,\text{rad/sample}$

Soln:

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)\, e^{j\omega n}\, d\omega.$$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha}\, e^{j\omega n}\, d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{+j\omega(n-\alpha)}\, d\omega. \quad (*)$$

$n-\alpha \neq 0$.

$$\frac{1}{2\pi}\left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\right]_{-\omega_c}^{\omega_c} \Rightarrow \frac{1}{2\pi j(n-\alpha)}\left[e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)}\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[2j\sin\omega_c(n-\alpha)\right] = \frac{\sin\omega_c(n-\alpha)}{\pi(n-\alpha)}$$

$n-\alpha = 0$, $(*)$ becomes,

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} 1\cdot d\omega = \frac{2\omega_c}{2\pi} = \frac{\omega_c}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin w_c(n-\alpha)}{\pi(n-\alpha)} , & n \neq \alpha \\ \dfrac{w_c}{\pi} , & n = \alpha . \end{cases}$$

as $w_c = 1$, to find $\alpha = ?$

w.r.t $\quad e^{-jw\alpha} = e^{-jw\left(\frac{N-1}{2}\right)}$ $\quad$ ($N = 7 \rightarrow$ given)

$$\frac{N-1}{2} = \alpha \Rightarrow \frac{7-1}{2} = \alpha \Rightarrow \boxed{\alpha = 3}$$

$$\therefore h_d(n) = \begin{cases} \dfrac{\sin(n-3)}{\pi(n-3)} , & n \neq 3 \\ \dfrac{w_c}{\pi} , & n = 3 . \end{cases}$$

as $N = 7$ $\{ n \Rightarrow 0 \rightarrow 6 \}$

n	$h_d(n)$	$w_H(n)$	$h(n) = h_d(n) \cdot w_H(n)$
0	0.01497 ~~7.486×10^-3~~	0	0
1	0.1447 ~~0.0723~~	0.25	0.0361
2	0.2678 ~~0.1339~~	0.75	0.20085
3	0.3183	1	0.3183
4	0.2678	0.75	0.20085
5	0.1447	0.25	0.0361
6	0.0149	0	0 .

$$w_H(n) = \begin{cases} \frac{1}{2}\left[1 - \cos\left(\frac{2\pi n}{M-1}\right)\right] , & n = 0 \rightarrow M-1 \\ 0 & \text{otherwise} \end{cases} \quad \text{for hanning window}$$

$\therefore h(n) = \{ 0, 0.0361, 0.20085, 0.3183, 0.20085, 0.0361, 0 \}$ and the $h(n)$ is symmetric.

Since N is odd and the $h(n)$ is symmetric

we have, $H(w) = \left[h\left(\frac{N-1}{2}\right) + \left[2 \sum_{n=0}^{N-3/2} h(n) \cos w\left(\frac{N-1}{2} - n\right) \right] e^{-jw\left(\frac{N-1}{2}\right)} \right]$

$$H(w) = h(3) + \{ 2h(0)\cos 3w + 2h(1)\cos 2w + 2h(2)\cos w \} e^{-j3w}.$$

$$H(w) = h(3) + \{ 2h(0)\cos 3w + 2h(1)\cos 2w + (2 \times 0.20085)\cos w \} e^{-j3w}$$

$$= 0.3183 + \{ (2\times 0)\cos 3w + (2 \times 0.0361)\cos 2w + (2\times 0.20085)\cos w \} e^{-j3w}$$

$$= 0.3183 + \{ 0.0722\cos 2w + 0.4016\cos w \} e^{-j3w}$$

5) Design an FIR low pass filter with frequency response $H_d(\omega) = \begin{cases} e^{-j\omega\frac{(N-1)}{2}} & -\pi/2 < \omega < \pi/2 \\ 0 & \text{otherwise} \end{cases}$ using rectangular window.

31/10/19

$e^{-j\omega x}$

Soln:

$h(n) = h_d(n)\, \omega_R(n)$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} H_d(\omega)\, e^{-j\omega n} . d\omega$$

$$\frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-j\omega\left(\frac{N-1}{2}\right)} . e^{j\omega n} . d\omega$$

$$\frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{j\omega\left[n - \left(\frac{N-1}{2}\right)\right]} . d\omega$$

Given $N = 7$

$$\frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{j\omega(n-3)} . d\omega \quad \longleftarrow (a)$$

case i) $n - 3 \neq 0 \Rightarrow n \neq 3$

$$\frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{j\omega(n-3)} . d\omega = \frac{1}{2\pi}\left[\frac{e^{j\omega(n-3)}}{j(n-3)}\right]$$

$$\frac{1}{2\pi j(n-3)} \left[e^{j\omega(n-3)} \right]_{-\pi/2}^{\pi/2}$$

$$\frac{1}{2\pi j(n-3)} \left[e^{j\pi/2(n-3)} - e^{-j\pi/2(n-3)} \right]$$

$$\frac{1}{2\pi j(n-3)} \left[2j \sin \pi/2 (n-3) \right] = \frac{\sin \pi/2 (n-3)}{\pi(n-3)}$$

case (ii). $n-3=0 \Rightarrow n=3$.

(*) becomes,

$$\frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1.dw = \frac{1}{2\pi}\left[\pi/2+\pi/2\right] = \frac{\pi}{2\pi} = \frac{1}{2}$$

$$h_d(n) = \begin{cases} \frac{\sin \pi/2 (n-3)}{\pi(n-3)} & n \neq 3. \\ 1/2 & n = 3. \end{cases}$$

as $N = 7$ $(\because n = 0 \to 6)$

n	$h_d(n)$	$w_R(n)$	$h(n) = h_d(n).w_R(n)$
0	-0.1061	1	-0.1061
1	0	1	0
2	0.318	1	0.318
3	0.5	1	0.5
4	0.318	1	0.318
5	0	1	0
6	-0.1061	1	-0.1061

as N is odd and symmetric,

$$H(w) = \left[h\left(\frac{N-1}{2}\right) + 2\sum_{n=0}^{\frac{N-3}{2}} h(n)\cos w\left(\frac{N-1}{2}-n\right)\right]e^{-jw\left(\frac{N-1}{2}\right)}$$

$$= h(3) + \left\{2h(0)\cos 3w + 2h(1)\cos 2w + 2h(2)\cos w\right\}e^{-j3w}.$$

$$= 0.5 + \left\{(2\times -0.1061)\cos 3w + (2\times 0)\cos 2w + (2\times 0.318)\cos w\right\}e^{-j3w}$$

$$H(w) = 0.5 + \left\{-0.212\cos 3w + 0 + 0.636\cos w\right\}e^{-j3w}$$

7. Design the bandpass linear phase FIR filter having cut-off frequencies $\omega_{c_1} = 1$ rad/sample, $\omega_{c_2} = 2$ rad/sample. Obtain the unit sample response through the following window.

$$w(n) = \begin{cases} 1 & 0 \le n \le 6 \\ 0 & \text{otherwise} \end{cases} \quad \text{\} rectangular pulse}$$

And also obtain frequency response.

Soln:

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)e^{j\omega n}\,d\omega. \rightarrow \text{desired frequency response of BPF}$$

$$h_d(n) = \frac{1}{2\pi}\left[\int_{-\omega_{c_2}}^{-\omega_{c_1}} H_d(\omega)\,e^{j\omega n} + \int_{\omega_{c_1}}^{\omega_{c_2}} H_d(\omega)\,e^{j\omega n}.d\omega\right] \rightarrow (*).$$

$$\frac{1}{2\pi}\left[\int_{-\omega_{c_2}}^{-\omega_{c_1}} e^{-j\omega\alpha}.e^{j\omega n}.d\omega + \int_{\omega_{c_1}}^{\omega_{c_2}} e^{-j\omega\alpha}.e^{j\omega n}.d\omega\right]$$

$$\frac{1}{2\pi}\left[\int_{-\omega_{c_2}}^{-\omega_{c_1}} e^{j\omega(n-\alpha)}.d\omega + \int_{\omega_{c_1}}^{\omega_{c_2}} e^{j\omega(n-\alpha)}.d\omega\right] \text{---} (*).$$

$n \neq \alpha$,

$$\frac{1}{2\pi}\left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\Bigg|_{-\omega_{c_2}}^{-\omega_{c_1}} + \frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\Bigg|_{\omega_{c_1}}^{\omega_2}\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[e^{-j\omega_{c_1}(n-\alpha)} - e^{-j\omega_{c_2}(n-\alpha)} + e^{j\omega_{c_2}(n-\alpha)} - e^{j\omega_{c_1}(n-\alpha)}\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[-2j\sin\omega_{c_1}(n-\alpha) + 2j\sin\omega_{c_2}(n-\alpha)\right]$$

$$\frac{1}{\pi(n-\alpha)}\left[\sin\omega_{c_2}(n-\alpha) - \sin\omega_{c_1}(n-\alpha)\right]$$

case (ii) $n = \alpha$.

(A) becomes,

$$h_d(n) = \frac{1}{2\pi}\left[\int_{-\omega_{c2}}^{-\omega_{c1}} e^0 . d\omega + \int_{\omega_{c1}}^{\omega_{c2}} e^0 . d\omega\right]$$

$$\frac{1}{2\pi}\left[-\omega_{c1} + \omega_{c2} + \omega_{c2} - \omega_{c1}\right]$$

$$\frac{1}{2\pi}\left[-2\omega_{c1} + 2\omega_{c2}\right] = \frac{\omega_{c2} - \omega_{c1}}{\pi}$$

$$h_d(n) = \begin{cases} \dfrac{\sin\omega_{c2}(n-\alpha) - \sin\omega_{c1}(n-\alpha)}{\pi(n-\alpha)} & n \neq \alpha \\[2ex] \dfrac{\omega_{c2} - \omega_{c1}}{\pi} & n = \alpha. \end{cases}$$

$\omega_{c2} = 2, \quad \omega_{c1} = 1.$

$$h_d(n) = \begin{cases} \dfrac{\sin 2(n-3) - \sin \cancel{1}(n-3)}{\pi(n-3)} & n \neq \alpha \\[2ex] 1/\pi & n = \alpha \end{cases}$$

$$e^{-j\omega\left(\frac{N-1}{2}\right)} = e^{-j\omega\alpha}$$

$$\alpha = \frac{N-1}{2} \Rightarrow \frac{\cancel{N}-1}{2} = \alpha \Rightarrow \boxed{\alpha = 3}$$

n	$h_d(n)$	$w_R(n)$	$h(n) = h_d(n) . w_R(n)$
0	-0.044	1	-0.044
1	-0.2651	1	-0.2651
2	0.0215	1	0.0215
3	$1/\pi$	1	0.3183
4	0.0215	1	0.0215
5	-0.2651	1	-0.2651
6	-0.044	1	0.044

$$h(n) = \{-0.04, -0.265, 0.025, 0.3183, 0.025, -0.265, -0.04\}$$

5) Design an highpass FIR filter using hamming window given $\omega_c = 1\,rad/sec$ and $N = 7$. Also find frequency response, magnitude & phase response. 7/11/19.

NOTE: ~~Desired~~ frequency response

Low pass filter:
$$H_d(\omega) = H_d(e^{-j\omega}) = \begin{cases} e^{-j\omega\alpha} & -\omega_c \le \omega \le \omega_c \\ 0 & \text{Otherwise.} \end{cases}$$

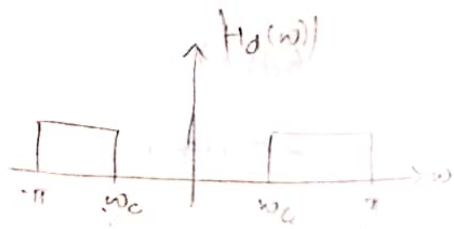
High pass filter:
$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & -\pi \le \omega \le -\omega_c,\ \omega_c \le \omega \le \pi \\ 0 & -\omega_c \le \omega \le \omega_c \end{cases}$$

Band pass filter:
$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & -\omega_{c_2} \le \omega \le -\omega_{c_1};\ \omega_{c_1} \le \omega \le \omega_{c_2} \\ 0 & \text{otherwise,} \end{cases}$$

Bandstop filter:
$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & -\pi \le \omega \le -\omega_{c_2};\ -\omega_{c_1} \le \omega \le \omega_{c_1} \ \&\ \omega_{c_2} \le \omega \le \pi \\ 0 & \text{Otherwise.} \end{cases}$$

$$\alpha = \frac{N-1}{2}$$

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)\, e^{j\omega n}\, d\omega.$$



$$h(n) = h_d(n)\, \omega_H(n)$$

Soln:
$$h_d(n) = \frac{1}{2\pi}\left[\int_{-\pi}^{-\omega_c} H_d(\omega).\, e^{j\omega n}.\, d\omega + \int_{\omega_c}^{\pi} H_d(\omega).\, e^{j\omega n}.\, d\omega\right]$$

$$hd(n) = \frac{1}{2\pi}\left[\int_{-\pi}^{-\omega_c} e^{-j\omega\alpha}\cdot e^{j\omega n}\cdot d\omega + \int_{\omega_c}^{\pi} e^{-j\omega\alpha}\cdot e^{j\omega n}\cdot d\omega\right] - (*).$$

$$\frac{1}{2\pi}\left[\int_{\pi}^{-\omega_c} e^{j\omega(n-\alpha)}\cdot d\omega + \int_{\omega_c}^{\pi} e^{j\omega(n-\alpha)}\cdot d\omega\right]$$

$$\frac{1}{2\pi}\left[\left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\right]_{-\pi}^{-\omega_c} + \left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)}\right]_{+\omega_c}^{\pi}\right]$$

$$\frac{1}{2\pi}\left\{\left[\frac{1}{j(n-\alpha)}\left(e^{-j\omega_c(n-\alpha)} - e^{-j\pi(n-\alpha)}\right)\right] + \frac{1}{j(n-\alpha)}\left(e^{j\pi(n-\alpha)} - e^{j\omega_c(n-\alpha)}\right)\right]\right\}$$

$$\frac{1}{2\pi j(n-\alpha)}\left[e^{-j\omega_c(n-\alpha)} - e^{-j\pi(n-\alpha)} + e^{j\pi(n-\alpha)} - e^{j\omega_c(n-\alpha)}\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[\left(e^{-j\omega_c(n-\alpha)} - e^{j\omega_c(n-\alpha)}\right) + \left(e^{j\pi(n-\alpha)} - e^{-j\pi(n-\alpha)}\right)\right]$$

$$\frac{1}{2\pi j(n-\alpha)}\left[-2j\sin\omega_c(n-\alpha) + 2j\sin\pi(n-\alpha)\right]$$

$$\frac{1}{\pi(n-\alpha)}\left[-\sin\omega_c(n-\alpha) + \sin\pi(n-\alpha)\right]$$

$$\frac{1}{\pi(n-\alpha)}\left[\sin\pi(n-\alpha) - \sin\omega_c(n-\alpha)\right]$$

ii) When $n = \alpha$, in (*).

$$hd(n) = \frac{1}{2\pi}\left[\int_{-\pi}^{-\omega_c} d\omega + \int_{\omega_c}^{\pi} d\omega\right]$$

$$\frac{1}{2\pi}\left[-\omega_c + \pi + \pi - \omega_c\right] = \frac{1}{2\pi}\left[2\pi - 2\omega_c\right]$$

$$hd(n) = \frac{\pi - \omega_c}{\pi}$$

$$\frac{\sin n\pi(n-3) - \sin(n-3)}{\pi(n-3)}$$

$$hd(n) = \begin{cases} \dfrac{\sin\pi(n-\alpha) - \sin\omega_c(n-\alpha)}{\pi(n-\alpha)} & n \neq \alpha \\[2ex] \dfrac{\pi-\omega_c}{\pi} & n = \alpha . \end{cases}$$

w.k.t, $h(n) = hd(n).\omega_H(n)$

$N = 7$

$w_H(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right).$

$\therefore \underline{\alpha = 3}$

n	hd(n)	$w_H(n)$	$h(n) = hd(n).w_H(n)$
0	-0.01497	0.08	-0.011976
1	-0.1447.	0.31	-0.044857
2	-0.2678	0.77	-0.20620
3	0,681	1	0.681
4	-0.2678	0.77	-0.20620
5	-0.1447	0.31	-0.04485
6	-0.01497	0.08	-0.011976.

h(n) is odd and symmetric,

$$H(\omega) = \left\{h\left(\frac{N-1}{2}\right) + 2\sum_{n=0}^{\frac{N-3}{2}} h[n]\cos\omega\left(\frac{N-1}{2}-n\right)\right\}e^{-j\omega\left(\frac{N-1}{2}\right)}.$$

$$\left\{h(3) + 2\sum_{n=0}^{2} h(n)\cos\omega(3-n)\right\}e^{-j3\omega}.$$

$$\left\{h(3) + 2h(0)\cos 3\omega + 2h(1)\cos 2\omega + 2h(2)\cos\omega\right\}e^{-j3\omega}.$$

~~-0.01197~~

$$\left\{1 - 0.0239\cos 3\omega - 0.0897\cos 2\omega - 0.4124\cos\omega\right\}e^{-j3\omega}$$

$$H(\omega) = |H(\omega)| e^{j\angle H(\omega)}.$$

$$|H(\omega)| = 1 - 0.0239\cos 3\omega - 0.0897\cos 2\omega - 0.4124\cos\omega$$

$$\angle H(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right) & H(\omega) \geq 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi & H(\omega) < 0. \end{cases} \qquad \begin{cases} -3\omega \\ -3\omega + \pi \end{cases}$$

9) Design an FIR filter (low pass) using rectangular window with pass band gain of 0dB, cut off frequency of 200Hz and sampling frequency of 1KHz. Assume length of impulse response as 7.

Soln: Given: $F_C = 200Hz$, $F_S = 1000Hz$. $n = 0 \ldots 6$; $N = 7$.

$$\left.\begin{matrix}\text{Normalizing} \\ \text{cut off freq}\end{matrix}\right\} = f_c = \frac{F_C}{F_S} = \frac{200}{1000} = 0.2 \text{ cycles/samples.}$$

$$\omega_c = 2\pi f_c = 2\pi \times 0.2 = 0.4\pi$$

For LPF

$$h(n) = \begin{cases} \frac{\sin\omega_c(n-\alpha)}{\pi(n-\alpha)} & n \neq \alpha \\ \omega_c/\pi & n = \alpha. \end{cases}$$

$$\alpha = \frac{N-1}{2} \Rightarrow \frac{7-1}{2} = 3.$$

$$h(n) = \begin{cases} \frac{\sin 0.4\pi(n-3)}{\pi(n-3)} & n \neq 3. \\ \frac{0.4\pi}{\pi} & n = 3. \end{cases}$$

$h(0) = \cancel{0.0302}$ -0.062

$h(1) = 0.0935$

$h(2) = 0.302$

$h(3) = 0.4$

$h(4) = 0.302$

$h(5) = 0.093$

$h(6) = -0.062$

impulse response: $h(n) =$

$\{-0.062, 0.0935, 0.302, 0.4, 0.302, 0.093, -0.062$

Note : Attenuation and Transition width of window

Window	Transition width of the main lobe	Minimum stop band attenuation	Relative amplitude of side lobe
Rectangular	$\frac{4\pi}{M+1}$	-21dB	-13dB
Bartlett	$\frac{8\pi}{M}$	-25dB	-25dB
Hanning	$\frac{8\pi}{M}$	-44dB	-31dB
Hamming	$\frac{8\pi}{M}$	-53dB	-41dB
Blackman	$\frac{12\pi}{M}$	-74dB	-57dB

8/11/19

Design low pass digital filter to be used in an A/D — H(Z) — D/A, structure that will have a -3dB cut off at $\Omega_c = 30\pi$ radians/sec. and an attenuation of 50 dB at 45π radians/sec. The filter is required to have a linear phase, and the system will use a sampling rate ~~weight~~ of 100 samples/sec.

Soln:

~~Desty~~ Given: $\Omega_c = 30\pi$ rad/sec, $F_s = 100Hz$. $A_s = 50dB$.

$\Omega_s = 45\pi$ rad/sec

To find the type of window.

Since attenuation is 50dB (refering to above table it corresponds to Hamming window (53dB).

To find M/N (order):

$$N = k\left[\frac{2\pi}{\omega_2 - \omega_1}\right]$$

main the width.

$$k\left(\frac{2\pi}{N}\right) = \left(\frac{8\pi}{N}\right) \quad \{\text{from table for hamming w/w}\}.$$

$$\underline{\underline{k = 4}}$$

Specification: $\omega_c = \frac{\Omega_c}{F_s} = \frac{30\pi}{100} = 0.3\pi$

$$\omega_1 = \frac{\Omega_1}{F_s} = \frac{30\pi}{100} = 0.3\pi$$

$$\omega_2 = \frac{\Omega_2}{F_s} = \frac{45\pi}{100} = 0.45\pi$$

$$N = 4\left[\frac{2\pi}{0.45\pi - 0.3\pi}\right] = 53.33 \simeq 55 \quad \{\text{round off to next odd no.}$$

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & -\omega_c \le \omega \le \omega_c \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} e^{-j\omega\left(\frac{N-1}{2}\right)} & -\omega_c \le \omega \le \omega_c \\ 0 & \text{otherwise} \end{cases}$$

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)e^{j\omega n}.d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{-j\omega\left(\frac{N-1}{2}\right)}.e^{j\omega n}.d\omega.$$

$$h_d(n) = \begin{cases} \dfrac{\sin\omega_c\left(n - \frac{N-1}{2}\right)}{\pi\left(n - \frac{N-1}{2}\right)}, & n \neq N/2 \\[2ex] \dfrac{\omega_c}{\pi} & n = \dfrac{N-1}{2} \end{cases}$$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{+j\omega\left(n - \left(\frac{N-1}{2}\right)\right)}.d\omega$$

on $N = 55$,

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n - 27)}.d\omega$$

case i) $n \neq 27$.

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-27)} d\omega =$$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega(n-2\alpha)} \cdot d\omega \quad \text{—} (*)$$

$$\frac{1}{2\pi}\left[\frac{e^{j\omega(n-2\alpha)}}{j(n-2\alpha)}\right]_{-\omega_c}^{\omega_c}$$

$$\frac{1}{2\pi}\left[\frac{e^{j\omega_c(n-2\alpha)} - e^{-j\omega_c(n-2\alpha)}}{j(n-2\alpha)}\right]$$

$$\frac{1}{2\pi j(n-2\alpha)}\left[2j \sin\omega_c(n-2\alpha)\right] = \frac{\sin\omega_c(n-2\alpha)}{\pi(n-2\alpha)}$$

at $n = 2\alpha$, $(*)$ becomes.

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{0} \cdot d\omega = \frac{1}{2\pi}(\omega_c + \omega_c) = \frac{\omega_c}{\pi}.$$

$$h_d(n) = \begin{cases} \dfrac{\sin\omega_c(n-2\alpha)}{\pi(n-2\alpha)}, & n \neq 2\alpha \\ \dfrac{\omega_c}{\pi}. & n = 2\alpha. \end{cases}$$

w.k.t, $h(n) = h_d(n) . \omega_H(n)$ $\{\omega_H(n) \rightarrow \text{hamming window}\}$

for $\omega_H(n) = \begin{cases} 0.54 - 0.46\cos\left(\dfrac{2\pi n}{N-1}\right) & 0 \dots M-1 \\ 0 & \text{otherwise.} \end{cases}$

$$\therefore h(n) = \left\{\frac{\sin\omega_c(n-2\alpha)}{\pi(n-2\alpha)}\right\}\left\{0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)\right\} +$$

$$\frac{\omega_c}{\pi}\left\{0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)\right\}.$$